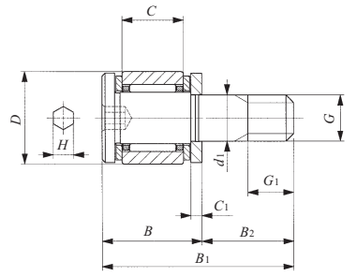
CAM FOLLOWERS

Thrust Disk Type Miniature Cam Followers With Cage/With Hexagon Hole



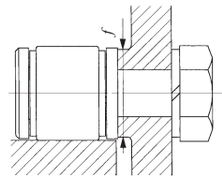


CFS··· W

Stud dia. 2 – 6mm

Stud dia. mm	Identification number	Mass (Ref.) g	Boundary dimensions mm					
			D	C	d_1	G	G_1	B
2	**CFS 2 W**	0.6	4.5	2.5	2	M2 ×0.4	2	4.5
2.5	**CFS 2.5 W**	1	5	3	2.5	M2.5 ×0.45	2.5	5
3	**CFS 3 W**	2	6	4	3	M3 ×0.5	3	6.5
4	**CFS 4 W**	4	8	5	4	M4 ×0.7	4	8
5	**CFS 5 W**	7	10	6	5	M5 ×0.8	5	9
6	**CFS 6 W**	13	12	7	6	M6 ×1	6	10.5

Remarks1. No oil hole is provided.
2. Provided with prepacked grease.



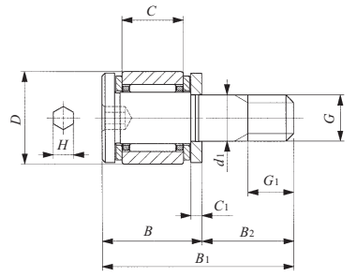
B_1	B_2	C_1	H	Mounting dimension f Min. mm	Maximum tightening torque N-cm	Basic dynamic load rating C N	Basic static load rating C_0 N	Maximum allowable static load N
8.5	4	0.7	0.9	4.3	9.1	288	202	194
10	5	0.7	0.9	4.8	18.7	428	351	313
12.5	6	0.7	1.3	5.8	33.5	629	611	399
16	8	1.0	1.5	7.7	77.7	1 120	1 120	785
19	10	1.0	2	9.6	158	1 570	1 850	1 370
22.5	12	1.2	2.5	11.6	268	2 090	2 200	1 920

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

CAM FOLLOWERS

Thrust Disk Type Miniature Cam Followers · Stainless Steel Made With Cage/With Hexagon Hole



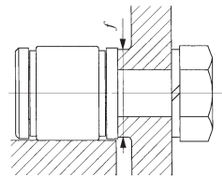


CFS···FW

Stud dia.2 – 6mm

Stud dia. mm	Identification number	Mass (Ref.) g	Boundary dimensions mm					
			D	C	d_1	G	G_1	B
2	**CFS 2 FW**	0.6	4.5	2.5	2	M2 ×0.4	2	4.5
2.5	**CFS 2.5 FW**	1	5	3	2.5	M2.5 ×0.45	2.5	5
3	**CFS 3 FW**	2	6	4	3	M3 ×0.5	3	6.5
4	**CFS 4 FW**	4	8	5	4	M4 ×0.7	4	8
5	**CFS 5 FW**	7	10	6	5	M5 ×0.8	5	9
6	**CFS 6 FW**	13	12	7	6	M6 ×1	6	10.5

Remarks1. No oil hole is provided.
2. Provided with prepacked grease.



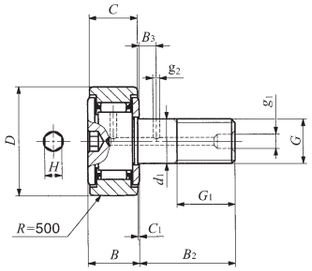
B_1	B_2	C_1	H	Mounting dimension f Min. mm	Maximum tightening torque N-cm	Basic dynamic load rating C N	Basic static load rating C_0 N	Maximum allowable static load N
8.5	4	0.7	0.9	4.3	9.1	230	161	161
10	5	0.7	0.9	4.8	18.7	342	281	281
12.5	6	0.7	1.3	5.8	33.5	504	488	399
16	8	1.0	1.5	7.7	77.7	897	894	785
19	10	1.0	2	9.6	158	1 250	1 480	1 370
22.5	12	1.2	2.5	11.6	268	1 670	1 760	1 760

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

CAM FOLLOWERS

Inch Series Cam Followers With Cage/With Hexagon Hole



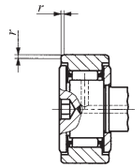


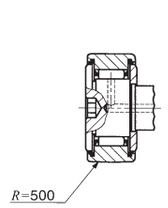
CR···BR

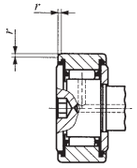
Stud dia. 4.826 — 22.225 mm

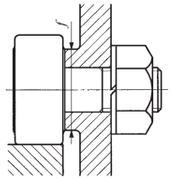
Stud dia. mm (inch)	Identification number Shield type With crowned outer ring	Shield type With cylindrical outer ring	Sealed type With crowned outer ring	Sealed type With cylindrical outer ring	Mass (Ref.) g	D	C	d_1	G UNF	G_1
4.826	CR 8 BR	CR 8 B	CR 8 BUUR	CR 8 BUU	9	12.700 (1/2)	8.731 (11/32)	4.826	No.10-32	6.350 (1/4)
	CR 8-1 BR	CR 8-1 B	CR 8-1 BUUR	CR 8-1 BUU	10	12.700 (1/2)	9.525 (3/8)	4.826	No.10-32	6.350 (1/4)
6.350 (1/4)	CR 10 BR	CR 10 B	CR 10 BUUR	CR 10 BUU	19	15.875 (5/8)	10.319 (13/32)	6.350 (1/4)	1/4 - 28	7.938 (5/16)
	CR 10-1 BR	CR 10-1 B	CR 10-1 BUUR	CR 10-1 BUU	21	15.875 (5/8)	11.112 (7/16)	6.350 (1/4)	1/4 - 28	7.938 (5/16)
9.525 (3/8)	CR 12 BR	CR 12 B	CR 12 BUUR	CR 12 BUU	35	19.050 (3/4)	12.700 (1/2)	9.525 (3/8)	3/8 - 24	9.525 (3/8)
	CR 14 BR	CR 14 B	CR 14 BUUR	CR 14 BUU	46	22.225 (7/8)	12.700 (1/2)	9.525 (3/8)	3/8 - 24	9.525 (3/8)
11.112 (7/16)	CR 16 BR	CR 16 B	CR 16 BUUR	CR 16 BUU	73	25.400 (1)	15.875 (5/8)	11.112 (7/16)	7/16 - 20	12.700 (1/2)
	CR 18 BR	CR 18 B	CR 18 BUUR	CR 18 BUU	88	28.575 (1 1/8)	15.875 (5/8)	11.112 (7/16)	7/16 - 20	12.700 (1/2)
12.700 (1/2)	CR 20 BR	CR 20 B	CR 20 BUUR	CR 20 BUU	132	31.750 (1 1/4)	19.050 (3/4)	12.700 (1/2)	1/2 - 20	15.875 (5/8)
	CR 22 BR	CR 22 B	CR 22 BUUR	CR 22 BUU	157	34.925 (1 3/8)	19.050 (3/4)	12.700 (1/2)	1/2 - 20	15.875 (5/8)
15.875 (5/8)	CR 24 BR	CR 24 B	CR 24 BUUR	CR 24 BUU	225	38.100 (1 1/2)	22.225 (7/8)	15.875 (5/8)	5/8 - 18	19.050 (3/4)
	CR 26 BR	CR 26 B	CR 26 BUUR	CR 26 BUU	260	41.275 (1 5/8)	22.225 (7/8)	15.875 (5/8)	5/8 - 18	19.050 (3/4)
19.050 (3/4)	CR 28 BR	CR 28 B	CR 28 BUUR	CR 28 BUU	365	44.450 (1 3/4)	25.400 (1)	19.050 (3/4)	3/4 - 16	22.225 (7/8)
	CR 30 BR	CR 30 B	CR 30 BUUR	CR 30 BUU	410	47.625 (1 7/8)	25.400 (1)	19.050 (3/4)	3/4 - 16	22.225 (7/8)
22.225 (7/8)	CR 32 BR	CR 32 B	CR 32 BUUR	CR 32 BUU	615	50.800 (2)	31.750 (1 1/4)	22.225 (7/8)	7/8 - 14	25.400 (1)
	CR 36 BR	CR 36 B	CR 36 BUUR	CR 36 BUU	750	57.150 (2 1/4)	31.750 (1 1/4)	22.225 (7/8)	7/8 - 14	25.400 (1)

Remarks1. Models with a stud diameter d_1 of 6.35 mm or less have no oil hole. Other models are provided with one oil hole each on the outside surface and end surface of the stud.
2. Provided with prepacked grease.









CR···B CR···BUUR CR···BUU

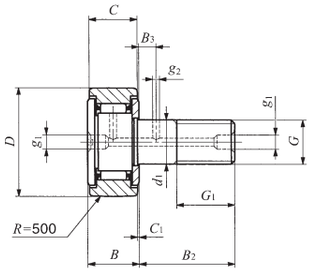
Boundary dimensions mm(inch) B max	B_2	B_3	C_1	g_1	g_2	H	r	Mounting dimension f Min. mm(inch)	Maximum tightening torque N-m	Basic dynamic load rating C N	Basic static load rating C_0 N
10.2(0.40)	12.700(1/2)	— (—)	0.794(1/32)	— (—)	— (—)	3.175(1/8)	0.397(1/64)	8.334(21/64)	1.4	2 520	2 140
10.9(0.43)	15.875(5/8)	— (—)	0.794(1/32)	— (—)	— (—)	3.175(1/8)	0.397(1/64)	8.334(21/64)	1.4	2 520	2 140
11.8(0.46)	15.875(5/8)	— (—)	0.794(1/32)	— (—)	— (—)	3.175(1/8)	0.397(1/64)	11.509(29/64)	3.4	3 650	3 670
12.5(0.49)	19.050(3/4)	— (—)	0.794(1/32)	— (—)	— (—)	3.175(1/8)	0.397(1/64)	11.509(29/64)	3.4	3 650	3 670
14.2(0.56)	22.225(7/8)	6.350(1/4)	0.794(1/32)	4.762(3/16)	2.381(3/32)	4.762(3/16)	0.794(1/32)	13.494(17/32)	10.8	4 420	5 110
14.2(0.56)	22.225(7/8)	6.350(1/4)	0.794(1/32)	4.762(3/16)	2.381(3/32)	4.762(3/16)	0.794(1/32)	15.081(19/32)	10.8	4 790	5 810
17.3(0.68)	25.400(1)	6.350(1/4)	0.794(1/32)	4.762(3/16)	3.175(1/8)	6.350(1/4)	1.191(3/64)	17.859(45/64)	17.4	8 810	10 800
17.3(0.68)	25.400(1)	6.350(1/4)	0.794(1/32)	4.762(3/16)	3.175(1/8)	6.350(1/4)	1.588(1/16)	19.050(3/4)	17.4	9 180	11 600
20.4(0.80)	31.750(1 1/4)	7.938(5/16)	0.794(1/32)	4.762(3/16)	3.175(1/8)	6.350(1/4)	1.588(1/16)	21.828(55/64)	27.7	14 200	16 000
20.4(0.80)	31.750(1 1/4)	7.938(5/16)	0.794(1/32)	4.762(3/16)	3.175(1/8)	6.350(1/4)	1.588(1/16)	21.828(55/64)	27.7	14 200	16 000
23.6(0.93)	38.100(1 1/2)	9.525(3/8)	0.794(1/32)	4.762(3/16)	3.969(5/32)	7.938(5/16)	1.588(1/16)	26.196(1 3/64)	55.7	18 600	24 300
23.6(0.93)	38.100(1 1/2)	9.525(3/8)	0.794(1/32)	4.762(3/16)	3.969(5/32)	7.938(5/16)	1.588(1/16)	26.196(1 3/64)	55.7	18 600	24 300
26.8(1.06)	44.450(1 3/4)	11.112(7/16)	0.794(1/32)	4.762(3/16)	3.969(5/32)	7.938(5/16)	1.588(1/16)	32.543(1 9/32)	100	25 100	38 200
26.8(1.06)	44.450(1 3/4)	11.112(7/16)	0.794(1/32)	4.762(3/16)	3.969(5/32)	7.938(5/16)	1.588(1/16)	32.543(1 9/32)	100	25 100	38 200
33.5(1.32)	50.800(2)	12.700(1/2)	0.794(1/32)	4.762(3/16)	4.762(3/16)	11.112(7/16)	1.588(1/16)	37.306(1 15/32)	162	32 500	63 900
33.5(1.32)	50.800(2)	12.700(1/2)	0.794(1/32)	4.762(3/16)	4.762(3/16)	11.112(7/16)	1.588(1/16)	37.306(1 15/32)	162	32 500	63 900

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

CAM FOLLOWERS

Inch Series Cam Followers With Cage/With Screwdriver Slot



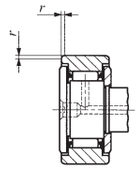


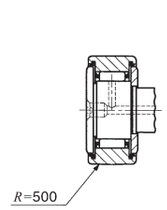
CR···R

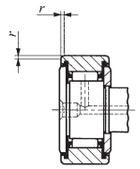
Stud dia. 4.826 — 22.225 mm

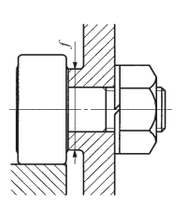
Stud dia. mm (inch)	Identification number: Shield type, With crowned outer ring	Shield type, With cylindrical outer ring	Sealed type, With crowned outer ring	Sealed type, With cylindrical outer ring	Mass (Ref.) g	D	C	d_1	G UNF	G_1
4.826	CR 8 R	CR 8	CR 8 UUR	CR 8 UU	9	12.700 (1/2)	8.731 (11/32)	4.826	No.10-32	6.350 (1/4)
	CR 8-1 R	CR 8-1	CR 8-1 UUR	CR 8-1 UU	10	12.700 (1/2)	9.525 (3/8)	4.826	No.10-32	6.350 (1/4)
6.350 (1/4)	CR 10 R	CR 10	CR 10 UUR	CR 10 UU	19	15.875 (5/8)	10.319 (13/32)	6.350 (1/4)	1/4 - 28	7.938 (5/16)
	CR 10-1 R	CR 10-1	CR 10-1 UUR	CR 10-1 UU	21	15.875 (5/8)	11.112 (7/16)	6.350 (1/4)	1/4 - 28	7.938 (5/16)
9.525 (3/8)	CR 12 R	CR 12	CR 12 UUR	CR 12 UU	35	19.050 (3/4)	12.700 (1/2)	9.525 (3/8)	3/8 - 24	9.525 (3/8)
	CR 14 R	CR 14	CR 14 UUR	CR 14 UU	46	22.225 (7/8)	12.700 (1/2)	9.525 (3/8)	3/8 - 24	9.525 (3/8)
11.112 (7/16)	CR 16 R	CR 16	CR 16 UUR	CR 16 UU	73	25.400 (1)	15.875 (5/8)	11.112 (7/16)	7/16 - 20	12.700 (1/2)
	CR 18 R	CR 18	CR 18 UUR	CR 18 UU	88	28.575 (1 1/8)	15.875 (5/8)	11.112 (7/16)	7/16 - 20	12.700 (1/2)
12.700 (1/2)	CR 20 R	CR 20	CR 20 UUR	CR 20 UU	132	31.750 (1 1/4)	19.050 (3/4)	12.700 (1/2)	1/2 - 20	15.875 (5/8)
	CR 22 R	CR 22	CR 22 UUR	CR 22 UU	157	34.925 (1 3/8)	19.050 (3/4)	12.700 (1/2)	1/2 - 20	15.875 (5/8)
15.875 (5/8)	CR 24 R	CR 24	CR 24 UUR	CR 24 UU	225	38.100 (1 1/2)	22.225 (7/8)	15.875 (5/8)	5/8 - 18	19.050 (3/4)
	CR 26 R	CR 26	CR 26 UUR	CR 26 UU	260	41.275 (1 5/8)	22.225 (7/8)	15.875 (5/8)	5/8 - 18	19.050 (3/4)
19.050 (3/4)	CR 28 R	CR 28	CR 28 UUR	CR 28 UU	365	44.450 (1 3/4)	25.400 (1)	19.050 (3/4)	3/4 - 16	22.225 (7/8)
	CR 30 R	CR 30	CR 30 UUR	CR 30 UU	410	47.625 (1 7/8)	25.400 (1)	19.050 (3/4)	3/4 - 16	22.225 (7/8)
22.225 (7/8)	CR 32 R	CR 32	CR 32 UUR	CR 32 UU	615	50.800 (2)	31.750 (1 1/4)	22.225 (7/8)	7/8 - 14	25.400 (1)
	CR 36 R	CR 36	CR 36 UUR	CR 36 UU	750	57.150 (2 1/4)	31.750 (1 1/4)	22.225 (7/8)	7/8 - 14	25.400 (1)

Remarks1. Models with a stud diameter d_1 of 6.35 mm or less (marked *) are provided with an oil hole on the stud head only. Other models are provided with one oil hole each on the head, outside surface and end surface of the stud.
2. Provided with prepacked grease.









CR CR···UUR CR···UU

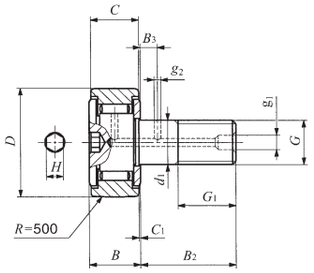
Boundary dimensions mm(inch): B max	B_2	B_3	C_1	g_1	g_2	r	Mounting dimension f Min. mm(inch)	Maximum tightening torque N-m	Basic dynamic load rating C N	Basic static load rating C_0 N
10.2(0.40)	12.700(1/2)	— (—)	0.794(1/32)	*3.175(1/8)	— (—)	0.397(1/64)	8.334(21/64)	1.4	2 520	2 140
10.9(0.43)	15.875(5/8)	— (—)	0.794(1/32)	*3.175(1/8)	— (—)	0.397(1/64)	8.334(21/64)	1.4	2 520	2 140
11.8(0.46)	15.875(5/8)	— (—)	0.794(1/32)	*3.175(1/8)	— (—)	0.397(1/64)	11.509(29/64)	3.4	3 650	3 670
12.5(0.49)	19.050(3/4)	— (—)	0.794(1/32)	*3.175(1/8)	— (—)	0.397(1/64)	11.509(29/64)	3.4	3 650	3 670
14.2(0.56)	22.225(7/8)	6.350(1/4)	0.794(1/32)	4.762(3/16)	2.381(3/32)	0.794(1/32)	13.494(17/32)	10.8	4 420	5 110
14.2(0.56)	22.225(7/8)	6.350(1/4)	0.794(1/32)	4.762(3/16)	2.381(3/32)	0.794(1/32)	15.081(19/32)	10.8	4 790	5 810
17.3(0.68)	25.400(1)	6.350(1/4)	0.794(1/32)	4.762(3/16)	3.175(1/8)	1.191(3/64)	17.859(45/64)	17.4	8 810	10 800
17.3(0.68)	25.400(1)	6.350(1/4)	0.794(1/32)	4.762(3/16)	3.175(1/8)	1.588(1/16)	19.050(3/4)	17.4	9 180	11 600
20.4(0.80)	31.750(1 1/4)	7.938(5/16)	0.794(1/32)	4.762(3/16)	3.175(1/8)	1.588(1/16)	21.828(55/64)	27.7	14 200	16 000
20.4(0.80)	31.750(1 1/4)	7.938(5/16)	0.794(1/32)	4.762(3/16)	3.175(1/8)	1.588(1/16)	21.828(55/64)	27.7	14 200	16 000
23.6(0.93)	38.100(1 1/2)	9.525(3/8)	0.794(1/32)	4.762(3/16)	3.969(5/32)	1.588(1/16)	26.196(1 3/64)	55.7	18 600	24 300
23.6(0.93)	38.100(1 1/2)	9.525(3/8)	0.794(1/32)	4.762(3/16)	3.969(5/32)	1.588(1/16)	26.196(1 3/64)	55.7	18 600	24 300
26.8(1.06)	44.450(1 3/4)	11.112(7/16)	0.794(1/32)	4.762(3/16)	3.969(5/32)	1.588(1/16)	32.543(1 9/32)	100	25 100	38 200
26.8(1.06)	44.450(1 3/4)	11.112(7/16)	0.794(1/32)	4.762(3/16)	3.969(5/32)	1.588(1/16)	32.543(1 9/32)	100	25 100	38 200
33.5(1.32)	50.800(2)	12.700(1/2)	0.794(1/32)	4.762(3/16)	4.762(3/16)	1.588(1/16)	37.306(1 15/32)	162	32 500	63 900
33.5(1.32)	50.800(2)	12.700(1/2)	0.794(1/32)	4.762(3/16)	4.762(3/16)	1.588(1/16)	37.306(1 15/32)	162	32 500	63 900

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

CAM FOLLOWERS

Inch Series Cam Followers Full Complement Type/With Hexagon Hole



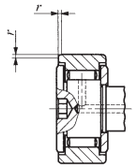


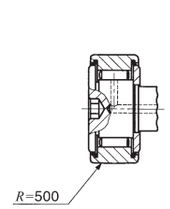
CR…VBR

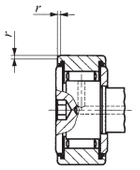
Stud dia. 4.826 — 22.225 mm

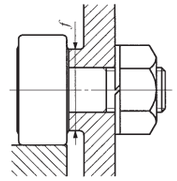
Stud dia. mm (inch)	Identification number Shield type With crowned outer ring	Shield type With cylindrical outer ring	Sealed type With crowned outer ring	Sealed type With cylindrical outer ring	Mass (Ref.) g	D	C	d_1	G UNF	G_1
4.826	CR 8 VBR	CR 8 VB	CR 8 VBUUR	CR 8 VBUU	9	12.700 (1/2)	8.731 (11/32)	4.826	No.10-32	6.350 (1/4)
	CR 8-1 VBR	CR 8-1VB	CR 8-1 VBUUR	CR 8-1 VBUU	10	12.700 (1/2)	9.525 (3/8)	4.826	No.10-32	6.350 (1/4)
6.350 (1/4)	CR 10 VBR	CR 10 VB	CR 10 VBUUR	CR 10 VBUU	19	15.875 (5/8)	10.319 (13/32)	6.350 (1/4)	1/4 - 28	7.938 (5/16)
	CR 10-1 VBR	CR 10-1VB	CR 10-1 VBUUR	CR 10-1 VBUU	21	15.875 (5/8)	11.112 (7/16)	6.350 (1/4)	1/4 - 28	7.938 (5/16)
9.525 (3/8)	CR 12 VBR	CR 12 VB	CR 12 VBUUR	CR 12 VBUU	36	19.050 (3/4)	12.700 (1/2)	9.525 (3/8)	3/8 - 24	9.525 (3/8)
	CR 14 VBR	CR 14 VB	CR 14 VBUUR	CR 14 VBUU	47	22.225 (7/8)	12.700 (1/2)	9.525 (3/8)	3/8 - 24	9.525 (3/8)
11.112 (7/16)	CR 16 VBR	CR 16 VB	CR 16 VBUUR	CR 16 VBUU	74	25.400 (1)	15.875 (5/8)	11.112 (7/16)	7/16 - 20	12.700 (1/2)
	CR 18 VBR	CR 18 VB	CR 18 VBUUR	CR 18 VBUU	85	28.575 (1 1/8)	15.875 (5/8)	11.112 (7/16)	7/16 - 20	12.700 (1/2)
12.700 (1/2)	CR 20 VBR	CR 20 VB	CR 20 VBUUR	CR 20 VBUU	137	31.750 (1 1/4)	19.050 (3/4)	12.700 (1/2)	1/2 - 20	15.875 (5/8)
	CR 22 VBR	CR 22 VB	CR 22 VBUUR	CR 22 VBUU	160	34.925 (1 3/8)	19.050 (3/4)	12.700 (1/2)	1/2 - 20	15.875 (5/8)
15.875 (5/8)	CR 24 VBR	CR 24 VB	CR 24 VBUUR	CR 24 VBUU	230	38.100 (1 1/2)	22.225 (7/8)	15.875 (5/8)	5/8 - 18	19.050 (3/4)
	CR 26 VBR	CR 26 VB	CR 26 VBUUR	CR 26 VBUU	265	41.275 (1 5/8)	22.225 (7/8)	15.875 (5/8)	5/8 - 18	19.050 (3/4)
19.050 (3/4)	CR 28 VBR	CR 28 VB	CR 28 VBUUR	CR 28 VBUU	372	44.450 (1 3/4)	25.400 (1)	19.050 (3/4)	3/4 - 16	22.225 (7/8)
	CR 30 VBR	CR 30 VB	CR 30 VBUUR	CR 30 VBUU	418	47.625 (1 7/8)	25.400 (1)	19.050 (3/4)	3/4 - 16	22.225 (7/8)
22.225 (7/8)	CR 32 VBR	CR 32 VB	CR 32 VBUUR	CR 32 VBUU	627	50.800 (2)	31.750 (1 1/4)	22.225 (7/8)	7/8 - 14	25.400 (1)
	CR 36 VBR	CR 36 VB	CR 36 VBUUR	CR 36 VBUU	759	57.150 (2 1/4)	31.750 (1 1/4)	22.225 (7/8)	7/8 - 14	25.400 (1)

Remarks1. Models with a stud diameter d_1 of 6.35 mm or less have no oil hole. Other models are provided with one oil hole each on the outside surface and end surface of the stud.
2. Provided with prepacked grease.









CR…VB CR…VBUUR CR…VBUU

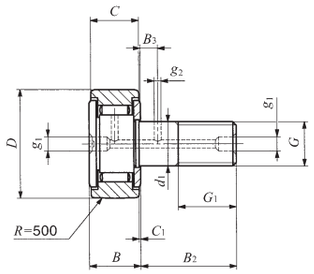
Boundary dimensions mm(inch) B max	B_2	B_3	C_1	g_1	g_2	H	r	Mounting dimension f Min. mm(inch)	Maximum tightening torque N-m	Basic dynamic load rating C N	Basic static load rating C_0 N
10.2(0.40)	12.700 (1/2)	— (—)	0.794(1/32)	— (—)	— (—)	3.175(1/8)	0.397(1/64)	8.334(21/64)	1.4	4 260	4 750
10.9(0.43)	15.875 (5/8)	— (—)	0.794(1/32)	— (—)	— (—)	3.175(1/8)	0.397(1/64)	8.334(21/64)	1.4	4 710	5 410
11.8(0.46)	15.875 (5/8)	— (—)	0.794(1/32)	— (—)	— (—)	3.175(1/8)	0.397(1/64)	11.509(29/64)	3.4	5 830	7 660
12.5(0.49)	19.050 (3/4)	— (—)	0.794(1/32)	— (—)	— (—)	3.175(1/8)	0.397(1/64)	11.509(29/64)	3.4	6 340	8 530
14.2(0.56)	22.225 (7/8)	6.350(1/4)	0.794(1/32)	4.762(3/16)	2.381(3/32)	4.762(3/16)	0.794(1/32)	13.494(17/32)	10.8	8 710	12 300
14.2(0.56)	22.225 (7/8)	6.350(1/4)	0.794(1/32)	4.762(3/16)	2.381(3/32)	4.762(3/16)	0.794(1/32)	15.081(19/32)	10.8	8 710	12 300
17.3(0.68)	25.400(1)	6.350(1/4)	0.794(1/32)	4.762(3/16)	3.175(1/8)	6.350(1/4)	1.191(3/64)	17.859(45/64)	17.4	13 100	22 700
17.3(0.68)	25.400(1)	6.350(1/4)	0.794(1/32)	4.762(3/16)	3.175(1/8)	6.350(1/4)	1.588(1/16)	19.050(3/4)	17.4	13 100	22 700
20.4(0.80)	31.750(1 1/4)	7.938(5/16)	0.794(1/32)	4.762(3/16)	3.175(1/8)	6.350(1/4)	1.588(1/16)	21.828(55/64)	27.7	23 600	31 700
20.4(0.80)	31.750(1 1/4)	7.938(5/16)	0.794(1/32)	4.762(3/16)	3.175(1/8)	6.350(1/4)	1.588(1/16)	21.828(55/64)	27.7	23 600	31 700
23.6(0.93)	38.100(1 1/2)	9.525(3/8)	0.794(1/32)	4.762(3/16)	3.969(5/32)	7.938(5/16)	1.588(1/16)	26.196(1 3/64)	55.7	28 200	40 100
23.6(0.93)	38.100(1 1/2)	9.525(3/8)	0.794(1/32)	4.762(3/16)	3.969(5/32)	7.938(5/16)	1.588(1/16)	26.196(1 3/64)	55.7	28 200	40 100
26.8(1.06)	44.450(1 3/4)	11.112(7/16)	0.794(1/32)	4.762(3/16)	3.969(5/32)	7.938(5/16)	1.588(1/16)	32.543(1 9/32)	100	35 300	55 600
26.8(1.06)	44.450(1 3/4)	11.112(7/16)	0.794(1/32)	4.762(3/16)	3.969(5/32)	7.938(5/16)	1.588(1/16)	32.543(1 9/32)	100	35 300	55 600
33.5(1.32)	50.800(2)	12.700(1/2)	0.794(1/32)	4.762(3/16)	4.762(3/16)	11.112 (7/16)	1.588(1/16)	37.306(1 15/32)	162	45 700	80 600
33.5(1.32)	50.800(2)	12.700(1/2)	0.794(1/32)	4.762(3/16)	4.762(3/16)	11.112 (7/16)	1.588(1/16)	37.306(1 15/32)	162	45 700	80 600

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

CAM FOLLOWERS

Inch Series Cam Followers Full Complement Type/With Screwdriver Slot



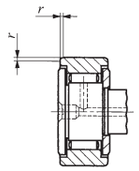


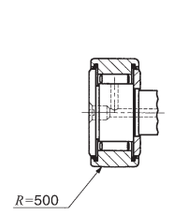
Stud dia. 4.826 — 31.750mm

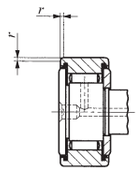
CR···VR

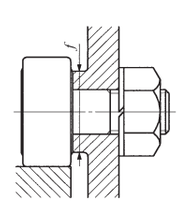
Stud dia. mm (inch)	Identification number Shield type With crowned outer ring	Shield type With cylindrical outer ring	Sealed type With crowned outer ring	Sealed type With cylindrical outer ring	Mass (Ref.) g	D	C	d_1	G UNF	G_1
4.826	CR 8 VR	CR 8 V	CR 8 VUUR	CR 8 VUU	9	12.700 (1/2)	8.731 (11/32)	4.826	No.10-32	6.350 (1/4)
	CR 8-1 VR	CR 8-1 V	CR 8-1 VUUR	CR 8-1 VUU	10	12.700 (1/2)	9.525 (3/8)	4.826	No.10-32	6.350 (1/4)
6.350 (1/4)	CR 10 VR	CR 10 V	CR 10 VUUR	CR 10 VUU	19	15.875 (5/8)	10.319 (13/32)	6.350 (1/4)	1/4 - 28	7.938 (5/16)
	CR 10-1 VR	CR 10-1 V	CR 10-1 VUUR	CR 10-1 VUU	21	15.875 (5/8)	11.112 (7/16)	6.350 (1/4)	1/4 - 28	7.938 (5/16)
9.525 (3/8)	CR 12 VR	CR 12 V	CR 12 VUUR	CR 12 VUU	36	19.050 (3/4)	12.700 (1/2)	9.525 (3/8)	3/8 - 24	9.525 (3/8)
	CR 14 VR	CR 14 V	CR 14 VUUR	CR 14 VUU	47	22.225 (7/8)	12.700 (1/2)	9.525 (3/8)	3/8 - 24	9.525 (3/8)
11.112 (7/16)	CR 16 VR	CR 16 V	CR 16 VUUR	CR 16 VUU	74	25.400 (1)	15.875 (5/8)	11.112 (7/16)	7/16 - 20	12.700 (1/2)
	CR 18 VR	CR 18 V	CR 18 VUUR	CR 18 VUU	85	28.575 (1 1/8)	15.875 (5/8)	11.112 (7/16)	7/16 - 20	12.700 (1/2)
12.700 (1/2)	CR 20 VR	CR 20 V	CR 20 VUUR	CR 20 VUU	137	31.750 (1 1/4)	19.050 (3/4)	12.700 (1/2)	1/2 - 20	15.875 (5/8)
	CR 22 VR	CR 22 V	CR 22 VUUR	CR 22 VUU	160	34.925 (1 3/8)	19.050 (3/4)	12.700 (1/2)	1/2 - 20	15.875 (5/8)
15.875 (5/8)	CR 24 VR	CR 24 V	CR 24 VUUR	CR 24 VUU	230	38.100 (1 1/2)	22.225 (7/8)	15.875 (5/8)	5/8 - 18	19.050 (3/4)
	CR 26 VR	CR 26 V	CR 26 VUUR	CR 26 VUU	265	41.275 (1 5/8)	22.225 (7/8)	15.875 (5/8)	5/8 - 18	19.050 (3/4)
19.050 (3/4)	CR 28 VR	CR 28 V	CR 28 VUUR	CR 28 VUU	372	44.450 (1 3/4)	25.400 (1)	19.050 (3/4)	3/4 - 16	22.225 (7/8)
	CR 30 VR	CR 30 V	CR 30 VUUR	CR 30 VUU	418	47.625 (1 7/8)	25.400 (1)	19.050 (3/4)	3/4 - 16	22.225 (7/8)
22.225 (7/8)	CR 32 VR	CR 32 V	CR 32 VUUR	CR 32 VUU	627	50.800 (2)	31.750 (1 1/4)	22.225 (7/8)	7/8 - 14	25.400 (1)
	CR 36 VR	CR 36 V	CR 36 VUUR	CR 36 VUU	759	57.150 (2 1/4)	31.750 (1 1/4)	22.225 (7/8)	7/8 - 14	25.400 (1)
31.750 (1 1/4)	—	—	—	CR 48 VUU	1 960	76.200 (3)	44.450 (1 3/4)	31.750 (1 1/4)	1 1/4 - 12	31.750 (1 1/4)

Remarks1. Models with a stud diameter d_1 of 6.35 mm or less (marked *) are provided with an oil hole on the stud head only. Other models are provided with one oil hole each on the head, outside surface and end surface of the stud.
2. Provided with prepacked grease.









CR···V CR···VUUR CR···VUU

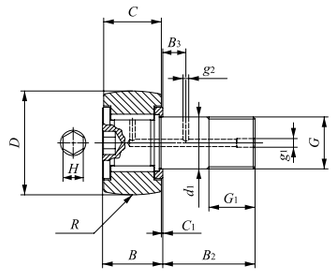
Boundary dimensions mm(inch) B max	B_2	B_3	C_1	g_1	g_2	r	Mounting dimension f Min. mm(inch)	Maximum tightening torque N-m	Basic dynamic load rating C N	Basic static load rating C_0 N
10.2(0.40)	12.700 (1/2)	— (—)	0.794 (1/32)	*3.175 (1/8)	— (—)	0.397 (1/64)	8.334 (21/64)	1.4	4 260	4 750
10.9(0.43)	15.875 (5/8)	— (—)	0.794 (1/32)	*3.175 (1/8)	— (—)	0.397 (1/64)	8.334 (21/64)	1.4	4 710	5 410
11.8(0.46)	15.875 (5/8)	— (—)	0.794 (1/32)	*3.175 (1/8)	— (—)	0.397 (1/64)	11.509 (29/64)	3.4	5 830	7 660
12.5(0.49)	19.050 (3/4)	— (—)	0.794 (1/32)	*3.175 (1/8)	— (—)	0.397 (1/64)	11.509 (29/64)	3.4	6 340	8 530
14.2(0.56)	22.225 (7/8)	6.350 (1/4)	0.794 (1/32)	4.762 (3/16)	2.381 (3/32)	0.794 (1/32)	13.494 (17/32)	10.8	8 710	12 300
14.2(0.56)	22.225 (7/8)	6.350 (1/4)	0.794 (1/32)	4.762 (3/16)	2.381 (3/32)	0.794 (1/32)	15.081 (19/32)	10.8	8 710	12 300
17.3(0.68)	25.400 (1)	6.350 (1/4)	0.794 (1/32)	4.762 (3/16)	3.175 (1/8)	1.191 (3/64)	17.859 (45/64)	17.4	13 100	22 700
17.3(0.68)	25.400 (1)	6.350 (1/4)	0.794 (1/32)	4.762 (3/16)	3.175 (1/8)	1.588 (1/16)	19.050 (3/4)	17.4	13 100	22 700
20.4(0.80)	31.750 (1 1/4)	7.938 (5/16)	0.794 (1/32)	4.762 (3/16)	3.175 (1/8)	1.588 (1/16)	21.828 (55/64)	27.7	23 600	31 700
20.4(0.80)	31.750 (1 1/4)	7.938 (5/16)	0.794 (1/32)	4.762 (3/16)	3.175 (1/8)	1.588 (1/16)	21.828 (55/64)	27.7	23 600	31 700
23.6(0.93)	38.100 (1 1/2)	9.525 (3/8)	0.794 (1/32)	4.762 (3/16)	3.969 (5/32)	1.588 (1/16)	26.196 (1 3/64)	55.7	28 200	40 100
23.6(0.93)	38.100 (1 1/2)	9.525 (3/8)	0.794 (1/32)	4.762 (3/16)	3.969 (5/32)	1.588 (1/16)	26.196 (1 3/64)	55.7	28 200	40 100
26.8(1.06)	44.450 (1 3/4)	11.112 (7/16)	0.794 (1/32)	4.762 (3/16)	3.969 (5/32)	1.588 (1/16)	32.543 (1 9/32)	100	35 300	55 600
26.8(1.06)	44.450 (1 3/4)	11.112 (7/16)	0.794 (1/32)	4.762 (3/16)	3.969 (5/32)	1.588 (1/16)	32.543 (1 9/32)	100	35 300	55 600
33.5(1.32)	50.800 (2)	12.700 (1/2)	0.794 (1/32)	4.762 (3/16)	4.762 (3/16)	1.588 (1/16)	37.306 (1 15/32)	162	45 700	80 600
33.5(1.32)	50.800 (2)	12.700 (1/2)	0.794 (1/32)	4.762 (3/16)	4.762 (3/16)	1.588 (1/16)	37.306 (1 15/32)	162	45 700	80 600
46.4(1.83)	63.500 (2 1/2)	15.875 (5/8)	1.588 (1/16)	6.350 (1/4)	4.762 (3/16)	2.381 (3/32)	51.991 (2 3/64)	500	77 600	172 000

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

CAM FOLLOWERS

Inch Series Heavy Duty Cam Followers Full Complement Type/With Hexagon Hole



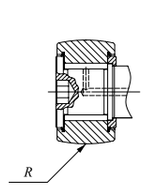


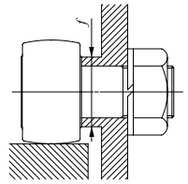
CRH···VBR

Stud dia. 6.350 — 50.800mm

Stud dia. mm (inch)	Identification number Shield type	Identification number Sealed type	Mass (Ref.) g	D	C	d_1	G UNF	G_1	B max
6.350 (1/4)	CRH 8-1 VBR	CRH 8-1 VBUUR	12	12.700 (1/2)	9.525 (3/8)	6.350 (1/4)	1/4 - 28	6.350 (1/4)	11.1(0.44)
	CRH 9 VBR	CRH 9 VBUUR	15	14.288 (9/16)	9.525 (3/8)	6.350 (1/4)	1/4 - 28	6.350 (1/4)	11.1(0.44)
7.938 (5/16)	CRH 10-1 VBR	CRH 10-1 VBUUR	23	15.875 (5/8)	11.112 (7/16)	7.938 (5/16)	5/16 - 24	7.938 (5/16)	12.8(0.50)
	CRH 11 VBR	CRH 11 VBUUR	27	17.462 (11/16)	11.112 (7/16)	7.938 (5/16)	5/16 - 24	7.938 (5/16)	12.8(0.50)
11.112 (7/16)	CRH 12 VBR	CRH 12 VBUUR	39	19.050 (3/4)	12.700 (1/2)	11.112 (7/16)	7/16 - 20	9.525 (3/8)	14.6(0.57)
	CRH 14 VBR	CRH 14 VBUUR	49	22.225 (7/8)	12.700 (1/2)	11.112 (7/16)	7/16 - 20	9.525 (3/8)	14.6(0.57)
15.875 (5/8)	CRH 16 VBR	CRH 16 VBUUR	93	25.400 (1)	15.875 (5/8)	15.875 (5/8)	5/8 - 18	12.700 (1/2)	17.9(0.70)
	CRH 18 VBR	CRH 18 VBUUR	109	28.575 (1 1/8)	15.875 (5/8)	15.875 (5/8)	5/8 - 18	12.700 (1/2)	17.9(0.70)
19.050 (3/4)	CRH 20 VBR	CRH 20 VBUUR	176	31.750 (1 1/4)	19.050 (3/4)	19.050 (3/4)	3/4 - 16	15.875 (5/8)	21.0(0.83)
	CRH 22 VBR	CRH 22 VBUUR	200	34.925 (1 3/8)	19.050 (3/4)	19.050 (3/4)	3/4 - 16	15.875 (5/8)	21.0(0.83)
22.225 (7/8)	CRH 24 VBR	CRH 24 VBUUR	296	38.100 (1 1/2)	22.225 (7/8)	22.225 (7/8)	7/8 - 14	19.050 (3/4)	24.3(0.96)
	CRH 26 VBR	CRH 26 VBUUR	329	41.275 (1 5/8)	22.225 (7/8)	22.225 (7/8)	7/8 - 14	19.050 (3/4)	24.3(0.96)
25.400 (1)	CRH 28 VBR	CRH 28 VBUUR	463	44.450 (1 3/4)	25.400 (1)	25.400 (1)	1 - 14 UNS	22.225 (7/8)	27.4(1.08)
	CRH 30 VBR	CRH 30 VBUUR	508	47.625 (1 7/8)	25.400 (1)	25.400 (1)	1 - 14 UNS	22.225 (7/8)	27.4(1.08)
28.575 (1 1/8)	CRH 32 VBR	CRH 32 VBUUR	722	50.800 (2)	31.750 (1 1/4)	28.575 (1 1/8)	1 1/8 - 12	25.400 (1)	34.2(1.35)
	CRH 36 VBR	CRH 36 VBUUR	858	57.150 (2 1/4)	31.750 (1 1/4)	28.575 (1 1/8)	1 1/8 - 12	25.400 (1)	34.2(1.35)
31.750 (1 1/4)	CRH 40 VBR	CRH 40 VBUUR	1 260	63.500 (2 1/2)	38.100 (1 1/2)	31.750 (1 1/4)	1 1/4 - 12	28.575 (1 1/8)	40.0(1.57)
	CRH 44 VBR	CRH 44 VBUUR	1 460	69.850 (2 3/4)	38.100 (1 1/2)	31.750 (1 1/4)	1 1/4 - 12	28.575 (1 1/8)	40.0(1.57)
38.100 (1 1/2)	CRH 48 VBR	CRH 48 VBUUR	2 100	76.200 (3)	44.450 (1 3/4)	38.100 (1 1/2)	1 1/2 - 12	31.750 (1 1/4)	46.4(1.83)
	CRH 52 VBR	CRH 52 VBUUR	2 380	82.550 (3 1/4)	44.450 (1 3/4)	38.100 (1 1/2)	1 1/2 - 12	31.750 (1 1/4)	46.4(1.83)
44.450 (1 3/4)	CRH 56 VBR	CRH 56 VBUUR	3 240	88.900 (3 1/2)	50.800 (2)	44.450 (1 3/4)	1 3/4 - 12UN	34.925 (1 3/8)	52.8(2.08)
50.800 (2)	CRH 64 VBR	CRH 64 VBUUR	4 960	101.600 (4)	57.150 (2 1/4)	50.800 (2)	2 - 12 UN	38.100 (1 1/2)	59.4(2.34)

Remarks1. Models with a stud diameter d_1 of 7.938 mm or less have no oil hole. Other models are provided with one oil hole each on the outside surface and end surface of the stud.
2. Provided with prepacked grease.





CRH···VBUUR

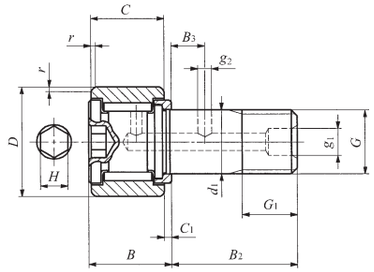
Boundary dimensions mm(inch) B_2	B_3	C_1	g_1	g_2	H	R	Mounting dimension f Min. mm(inch)	Maximum tightening torque N-m	Basic dynamic load rating C N	Basic static load rating C_0 N
15.875(5/8)	— (—)	0.794(1/32)	— (—)	— (—)	3.175(1/8)	180(7)	8.334(21/64)	3.4	4 710	5 410
15.875(5/8)	— (—)	0.794(1/32)	— (—)	— (—)	3.175(1/8)	180(7)	8.334(21/64)	3.4	4 710	5 410
19.050(3/4)	— (—)	0.794(1/32)	— (—)	— (—)	3.175(1/8)	200(8)	11.112(7/16)	6.8	6 340	8 530
19.050(3/4)	— (—)	0.794(1/32)	— (—)	— (—)	3.175(1/8)	200(8)	11.112(7/16)	6.8	6 340	8 530
22.225(7/8)	6.350(1/4)	0.794(1/32)	4.762(3/16)	2.381(3/32)	4.762(3/16)	250(10)	13.494(17/32)	17.6	8 710	12 300
22.225(7/8)	6.350(1/4)	0.794(1/32)	4.762(3/16)	2.381(3/32)	4.762(3/16)	250(10)	13.494(17/32)	17.6	8 710	12 300
25.400(1)	6.350(1/4)	1.588(1/16)	4.762(3/16)	2.381(3/32)	6.350(1/4)	300(12)	18.256(23/32)	57.8	13 100	22 700
25.400(1)	6.350(1/4)	1.588(1/16)	4.762(3/16)	2.381(3/32)	6.350(1/4)	300(12)	18.256(23/32)	57.8	13 100	22 700
31.750(1 1/4)	7.938(5/16)	1.588(1/16)	4.762(3/16)	2.381(3/32)	6.350(1/4)	360(14)	24.209(61/64)	103	23 600	31 700
31.750(1 1/4)	7.938(5/16)	1.588(1/16)	4.762(3/16)	2.381(3/32)	6.350(1/4)	360(14)	24.209(61/64)	103	23 600	31 700
38.100(1 1/2)	9.525(3/8)	1.588(1/16)	4.762(3/16)	2.381(3/32)	7.938(5/16)	500(20)	26.988(1 1/16)	162	28 200	40 100
38.100(1 1/2)	9.525(3/8)	1.588(1/16)	4.762(3/16)	2.381(3/32)	7.938(5/16)	500(20)	26.988(1 1/16)	162	28 200	40 100
44.450(1 3/4)	11.112(7/16)	1.588(1/16)	4.762(3/16)	2.381(3/32)	7.938(5/16)	500(20)	32.941(1 19/64)	258	35 300	55 600
44.450(1 3/4)	11.112(7/16)	1.588(1/16)	4.762(3/16)	2.381(3/32)	7.938(5/16)	500(20)	32.941(1 19/64)	258	35 300	55 600
50.800(2)	12.700(1/2)	1.588(1/16)	4.762(3/16)	3.175(1/8)	11.112(7/16)	600(24)	37.306(1 15/32)	356	45 700	80 600
50.800(2)	12.700(1/2)	1.588(1/16)	4.762(3/16)	3.175(1/8)	11.112(7/16)	600(24)	37.306(1 15/32)	356	45 700	80 600
57.150(2 1/4)	14.288(9/16)	1.588(1/16)	4.762(3/16)	3.175(1/8)	12.700(1/2)	760(30)	40.878(1 39/64)	500	61 400	116 000
57.150(2 1/4)	14.288(9/16)	1.588(1/16)	4.762(3/16)	3.175(1/8)	12.700(1/2)	760(30)	40.878(1 39/64)	500	61 400	116 000
63.500(2 1/2)	15.875(5/8)	1.588(1/16)	6.350(1/4)	3.175(1/8)	19.050(3/4)	760(30)	51.991(2 3/64)	892	77 600	172 000
63.500(2 1/2)	15.875(5/8)	1.588(1/16)	6.350(1/4)	3.175(1/8)	19.050(3/4)	760(30)	51.991(2 3/64)	892	77 600	172 000
69.850(2 3/4)	17.462(11/16)	1.588(1/16)	6.350(1/4)	3.175(1/8)	19.050(3/4)	760(30)	59.928(2 23/64)	1 450	111 000	239 000
88.900(3 1/2)	19.050(3/4)	1.588(1/16)	6.350(1/4)	3.175(1/8)	19.050(3/4)	760(30)	64.691(2 35/64)	2 190	142 000	317 000

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

CAM FOLLOWERS

Inch Series Heavy Duty Cam Followers Full Complement Type/With Hexagon Hole



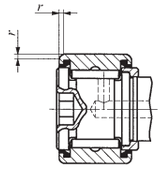


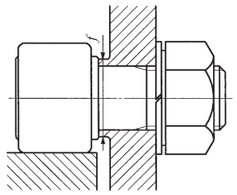
Stud dia. 6.350 — 50.800mm

CRH···VB

Stud dia. mm (inch)	Identification number Shield type	Identification number Sealed type	Mass (Ref.) g	D	C	d_1	G UNF	G_1	B max
6.350 (1/4)	CRH 8-1 VB	CRH 8-1 VBUU	12	12.700 (1/2)	9.525 (3/8)	6.350 (1/4)	1/4 - 28	6.350 (1/4)	11.1(0.44)
	CRH 9 VB	CRH 9 VBUU	15	14.288 (9/16)	9.525 (3/8)	6.350 (1/4)	1/4 - 28	6.350 (1/4)	11.1(0.44)
7.938 (5/16)	CRH 10-1 VB	CRH 10-1 VBUU	23	15.875 (5/8)	11.112 (7/16)	7.938 (5/16)	5/16 - 24	7.938 (5/16)	12.8(0.50)
	CRH 11 VB	CRH 11 VBUU	27	17.462 (11/16)	11.112 (7/16)	7.938 (5/16)	5/16 - 24	7.938 (5/16)	12.8(0.50)
11.112 (7/16)	CRH 12 VB	CRH 12 VBUU	39	19.050 (3/4)	12.700 (1/2)	11.112 (7/16)	7/16 - 20	9.525 (3/8)	14.6(0.57)
	CRH 14 VB	CRH 14 VBUU	49	22.225 (7/8)	12.700 (1/2)	11.112 (7/16)	7/16 - 20	9.525 (3/8)	14.6(0.57)
15.875 (5/8)	CRH 16 VB	CRH 16 VBUU	93	25.400 (1)	15.875 (5/8)	15.875 (5/8)	5/8 - 18	12.700 (1/2)	17.9(0.70)
	CRH 18 VB	CRH 18 VBUU	109	28.575 (1 1/8)	15.875 (5/8)	15.875 (5/8)	5/8 - 18	12.700 (1/2)	17.9(0.70)
19.050 (3/4)	CRH 20 VB	CRH 20 VBUU	176	31.750 (1 1/4)	19.050 (3/4)	19.050 (3/4)	3/4 - 16	15.875 (5/8)	21.0(0.83)
	CRH 22 VB	CRH 22 VBUU	200	34.925 (1 3/8)	19.050 (3/4)	19.050 (3/4)	3/4 - 16	15.875 (5/8)	21.0(0.83)
22.225 (7/8)	CRH 24 VB	CRH 24 VBUU	296	38.100 (1 1/2)	22.225 (7/8)	22.225 (7/8)	7/8 - 14	19.050 (3/4)	24.3(0.96)
	CRH 26 VB	CRH 26 VBUU	329	41.275 (1 5/8)	22.225 (7/8)	22.225 (7/8)	7/8 - 14	19.050 (3/4)	24.3(0.96)
25.400 (1)	CRH 28 VB	CRH 28 VBUU	463	44.450 (1 3/4)	25.400 (1)	25.400 (1)	1 - 14 UNS	22.225 (7/8)	27.4(1.08)
	CRH 30 VB	CRH 30 VBUU	508	47.625 (1 7/8)	25.400 (1)	25.400 (1)	1 - 14 UNS	22.225 (7/8)	27.4(1.08)
28.575 (1 1/8)	CRH 32 VB	CRH 32 VBUU	722	50.800 (2)	31.750 (1 1/4)	28.575 (1 1/8)	1 1/8 - 12	25.400 (1)	34.2(1.35)
	CRH 36 VB	CRH 36 VBUU	858	57.150 (2 1/4)	31.750 (1 1/4)	28.575 (1 1/8)	1 1/8 - 12	25.400 (1)	34.2(1.35)
31.750 (1 1/4)	CRH 40 VB	CRH 40 VBUU	1 260	63.500 (2 1/2)	38.100 (1 1/2)	31.750 (1 1/4)	1 1/4 - 12	28.575 (1 1/8)	40.0(1.57)
	CRH 44 VB	CRH 44 VBUU	1 460	69.850 (2 3/4)	38.100 (1 1/2)	31.750 (1 1/4)	1 1/4 - 12	28.575 (1 1/8)	40.0(1.57)
38.100 (1 1/2)	CRH 48 VB	CRH 48 VBUU	2 100	76.200 (3)	44.450 (1 3/4)	38.100 (1 1/2)	1 1/2 - 12	31.750 (1 1/4)	46.4(1.83)
	CRH 52 VB	CRH 52 VBUU	2 380	82.550 (3 1/4)	44.450 (1 3/4)	38.100 (1 1/2)	1 1/2 - 12	31.750 (1 1/4)	46.4(1.83)
44.450 (1 3/4)	CRH 56 VB	CRH 56 VBUU	3 240	88.900 (3 1/2)	50.800 (2)	44.450 (1 3/4)	1 3/4 - 12 UN	34.925 (1 3/8)	52.8(2.08)
50.800 (2)	CRH 64 VB	CRH 64 VBUU	4 960	101.600 (4)	57.150 (2 1/4)	50.800 (2)	2- 12 UN	38.100 (1 1/2)	59.4(2.34)

Remarks1. Models with a stud diameter d_1 of 7.938 mm or less have no oil hole. Other models are provided with one oil hole each on the outside surface and end surface of the stud.
2. Provided with prepacked grease.





CRH···VBUU

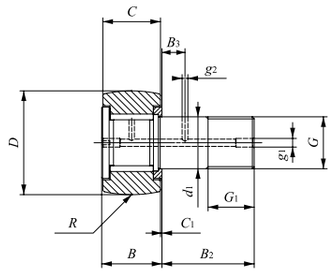
Boundary dimensions mm(inch) B_2	B_3	C_1	g_1	g_2	H	r	Mounting dimension f Min. mm(inch)	Maximum tightening torque N-m	Basic dynamic load rating C N	Basic static load rating C_0 N
15.875(5/8)	— (—)	0.794(1/32)	— (—)	— (—)	3.175(1/8)	0.397(1/64)	8.334(21/64)	3.4	4 710	5 410
15.875(5/8)	— (—)	0.794(1/32)	— (—)	— (—)	3.175(1/8)	0.397(1/64)	8.334(21/64)	3.4	4 710	5 410
19.050(3/4)	— (—)	0.794(1/32)	— (—)	— (—)	3.175(1/8)	0.397(1/64)	11.112 (7/16)	6.8	6 340	8 530
19.050(3/4)	— (—)	0.794(1/32)	— (—)	— (—)	3.175(1/8)	0.397(1/64)	11.112 (7/16)	6.8	6 340	8 530
22.225(7/8)	6.350(1/4)	0.794(1/32)	4.762(3/16)	2.381(3/32)	4.762(3/16)	0.794(1/32)	13.494(17/32)	17.6	8 710	12 300
22.225(7/8)	6.350(1/4)	0.794(1/32)	4.762(3/16)	2.381(3/32)	4.762(3/16)	0.794(1/32)	13.494(17/32)	17.6	8 710	12 300
25.400(1)	6.350(1/4)	1.588(1/16)	4.762(3/16)	2.381(3/32)	6.350(1/4)	1.191(3/64)	18.256(23/32)	57.8	13 100	22 700
25.400(1)	6.350(1/4)	1.588(1/16)	4.762(3/16)	2.381(3/32)	6.350(1/4)	1.588(1/16)	18.256(23/32)	57.8	13 100	22 700
31.750(1 1/4)	7.938(5/16)	1.588(1/16)	4.762(3/16)	2.381(3/32)	6.350(1/4)	1.588(1/16)	24.209(61/64)	103	23 600	31 700
31.750(1 1/4)	7.938(5/16)	1.588(1/16)	4.762(3/16)	2.381(3/32)	6.350(1/4)	1.588(1/16)	24.209(61/64)	103	23 600	31 700
38.100(1 1/2)	9.525(3/8)	1.588(1/16)	4.762(3/16)	2.381(3/32)	7.938(5/16)	1.588(1/16)	26.988(1 1/16)	162	28 200	40 100
38.100(1 1/2)	9.525(3/8)	1.588(1/16)	4.762(3/16)	2.381(3/32)	7.938(5/16)	1.588(1/16)	26.988(1 1/16)	162	28 200	40 100
44.450(1 3/4)	11.112(7/16)	1.588(1/16)	4.762(3/16)	2.381(3/32)	7.938(5/16)	1.588(1/16)	32.941(1 19/64)	258	35 300	55 600
44.450(1 3/4)	11.112(7/16)	1.588(1/16)	4.762(3/16)	2.381(3/32)	7.938(5/16)	1.588(1/16)	32.941(1 19/64)	258	35 300	55 600
50.800(2)	12.700(1/2)	1.588(1/16)	4.762(3/16)	3.175(1/8)	11.112(7/16)	1.588(1/16)	37.306(1 15/32)	356	45 700	80 600
50.800(2)	12.700(1/2)	1.588(1/16)	4.762(3/16)	3.175(1/8)	11.112(7/16)	1.588(1/16)	37.306(1 15/32)	356	45 700	80 600
57.150(2 1/4)	14.288(9/16)	1.588(1/16)	4.762(3/16)	3.175(1/8)	12.700(1/2)	2.381(3/32)	40.878(1 39/64)	500	61 400	116 000
57.150(2 1/4)	14.288(9/16)	1.588(1/16)	4.762(3/16)	3.175(1/8)	12.700(1/2)	2.381(3/32)	40.878(1 39/64)	500	61 400	116 000
63.500(2 1/2)	15.875(5/8)	1.588(1/16)	6.350(1/4)	3.175(1/8)	19.050(3/4)	2.381(3/32)	51.991(2 3/64)	892	77 600	172 000
63.500(2 1/2)	15.875(5/8)	1.588(1/16)	6.350(1/4)	3.175(1/8)	19.050(3/4)	2.381(3/32)	51.991(2 3/64)	892	77 600	172 000
69.850(2 3/4)	17.462(11/16)	1.588(1/16)	6.350(1/4)	3.175(1/8)	19.050(3/4)	2.381(3/32)	59.928(2 23/64)	1 450	111 000	239 000
88.900(3 1/2)	19.050(3/4)	1.588(1/16)	6.350(1/4)	3.175(1/8)	19.050(3/4)	2.381(3/32)	64.691(2 35/64)	2 190	142 000	317 000

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

CAM FOLLOWERS

Inch Series Heavy Duty Cam Followers Full Complement Type/With Screwdriver Slot



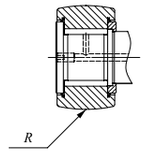


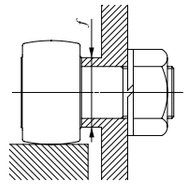
CRH···VR

Stud dia. 6.350 — 50.800mm

Stud dia. mm (inch)	Identification number Shield type	Identification number Sealed type	Mass (Ref.) g	D	C	d_1	G UNF	G_1	B max
6.350 (1/4)	CRH 8-1 VR	CRH 8-1 VUUR	12	12.700 (1/2)	9.525 (3/8)	6.350 (1/4)	1/4 - 28	6.350 (1/4)	11.1(0.44)
	CRH 9 VR	CRH 9 VUUR	15	14.288 (9/16)	9.525 (3/8)	6.350 (1/4)	1/4 - 28	6.350 (1/4)	11.1(0.44)
7.938 (5/16)	CRH 10-1 VR	CRH 10-1 VUUR	23	15.875 (5/8)	11.112 (7/16)	7.938 (5/16)	5/16 - 24	7.938 (5/16)	12.8(0.50)
	CRH 11 VR	CRH 11 VUUR	27	17.462 (11/16)	11.112 (7/16)	7.938 (5/16)	5/16 - 24	7.938 (5/16)	12.8(0.50)
11.112 (7/16)	CRH 12 VR	CRH 12 VUUR	39	19.050 (3/4)	12.700 (1/2)	11.112 (7/16)	7/16 - 20	9.525 (3/8)	14.6(0.57)
	CRH 14 VR	CRH 14 VUUR	49	22.225 (7/8)	12.700 (1/2)	11.112 (7/16)	7/16 - 20	9.525 (3/8)	14.6(0.57)
15.875 (5/8)	CRH 16 VR	CRH 16 VUUR	93	25.400 (1)	15.875 (5/8)	15.875 (5/8)	5/8 - 18	12.700 (1/2)	17.9(0.70)
	CRH 18 VR	CRH 18 VUUR	109	28.575 (1 1/8)	15.875 (5/8)	15.875 (5/8)	5/8 - 18	12.700 (1/2)	17.9(0.70)
19.050 (3/4)	CRH 20 VR	CRH 20 VUUR	176	31.750 (1 1/4)	19.050 (3/4)	19.050 (3/4)	3/4 - 16	15.875 (5/8)	21.0(0.83)
	CRH 22 VR	CRH 22 VUUR	200	34.925 (1 3/8)	19.050 (3/4)	19.050 (3/4)	3/4 - 16	15.875 (5/8)	21.0(0.83)
22.225 (7/8)	CRH 24 VR	CRH 24 VUUR	296	38.100 (1 1/2)	22.225 (7/8)	22.225 (7/8)	7/8 - 14	19.050 (3/4)	24.3(0.96)
	CRH 26 VR	CRH 26 VUUR	329	41.275 (1 5/8)	22.225 (7/8)	22.225 (7/8)	7/8 - 14	19.050 (3/4)	24.3(0.96)
25.400 (1)	CRH 28 VR	CRH 28 VUUR	463	44.450 (1 3/4)	25.400 (1)	25.400 (1)	1 - 14 UNS	22.225 (7/8)	27.4(1.08)
	CRH 30 VR	CRH 30 VUUR	508	47.625 (1 7/8)	25.400 (1)	25.400 (1)	1 - 14 UNS	22.225 (7/8)	27.4(1.08)
28.575 (1 1/8)	CRH 32 VR	CRH 32 VUUR	722	50.800 (2)	31.750 (1 1/4)	28.575 (1 1/8)	1 1/8 - 12	25.400 (1)	34.2(1.35)
	CRH 36 VR	CRH 36 VUUR	858	57.150 (2 1/4)	31.750 (1 1/4)	28.575 (1 1/8)	1 1/8 - 12	25.400 (1)	34.2(1.35)
31.750 (1 1/4)	CRH 40 VR	CRH 40 VUUR	1 260	63.500 (2 1/2)	38.100 (1 1/2)	31.750 (1 1/4)	1 1/4 - 12	28.575 (1 1/8)	40.0(1.57)
	CRH 44 VR	CRH 44 VUUR	1 460	69.850 (2 3/4)	38.100 (1 1/2)	31.750 (1 1/4)	1 1/4 - 12	28.575 (1 1/8)	40.0(1.57)
38.100 (1 1/2)	CRH 48 VR	CRH 48 VUUR	2 100	76.200 (3)	44.450 (1 3/4)	38.100 (1 1/2)	1 1/2 - 12	31.750 (1 1/4)	46.4(1.83)
	CRH 52 VR	CRH 52 VUUR	2 380	82.550 (3 1/4)	44.450 (1 3/4)	38.100 (1 1/2)	1 1/2 - 12	31.750 (1 1/4)	46.4(1.83)
44.450 (1 3/4)	CRH 56 VR	CRH 56 VUUR	3 240	88.900 (3 1/2)	50.800 (2)	44.450 (1 3/4)	1 3/4 - 12 UN	34.925 (1 3/8)	52.8(2.08)
50.800 (2)	CRH 64 VR	CRH 64 VUUR	4 960	101.600 (4)	57.150 (2 1/4)	50.800 (2)	2 - 12 UN	38.100 (1 1/2)	59.4(2.34)

Remarks1. Models with a stud diameter d_1 of 7.938 mm or less (marked *) are provided with an oil hole on the stud head only. Other models are provided with one oil hole each on the head, outside surface and end surface of the stud.
2. Provided with prepacked grease.





CRH···VUUR

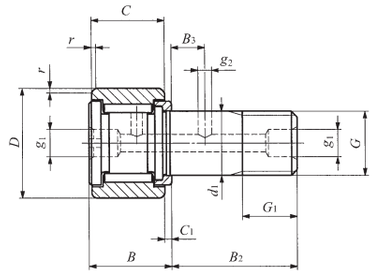
Boundary dimensions mm(inch) B_2	B_3	C_1	g_1	g_2	R	Mounting dimension f Min. mm(inch)	Maximum tightening torque N-m	Basic dynamic load rating C N	Basic static load rating C_0 N
15.875 (5/8)	— (—)	0.794 (1/32)	*3.175 (1/8)	— (—)	180 (7)	8.334 (21/64)	3.4	4 710	5 410
15.875 (5/8)	— (—)	0.794 (1/32)	*3.175 (1/8)	— (—)	180 (7)	8.334 (21/64)	3.4	4 710	5 410
19.050 (3/4)	— (—)	0.794 (1/32)	*3.175 (1/8)	— (—)	200 (8)	11.112 (7/16)	6.8	6 340	8 530
19.050 (3/4)	— (—)	0.794 (1/32)	*3.175 (1/8)	— (—)	200 (8)	11.112 (7/16)	6.8	6 340	8 530
22.225 (7/8)	6.350 (1/4)	0.794 (1/32)	4.762 (3/16)	2.381 (3/32)	250 (10)	13.494 (17/32)	17.6	8 710	12 300
22.225 (7/8)	6.350 (1/4)	0.794 (1/32)	4.762 (3/16)	2.381 (3/32)	250 (10)	13.494 (17/32)	17.6	8 710	12 300
25.400 (1)	6.350 (1/4)	1.588 (1/16)	4.762 (3/16)	2.381 (3/32)	300 (12)	18.256 (23/32)	57.8	13 100	22 700
25.400 (1)	6.350 (1/4)	1.588 (1/16)	4.762 (3/16)	2.381 (3/32)	300 (12)	18.256 (23/32)	57.8	13 100	22 700
31.750 (1 1/4)	7.938 (5/16)	1.588 (1/16)	4.762 (3/16)	2.381 (3/32)	360 (14)	24.209 (61/64)	103	23 600	31 700
31.750 (1 1/4)	7.938 (5/16)	1.588 (1/16)	4.762 (3/16)	2.381 (3/32)	360 (14)	24.209 (61/64)	103	23 600	31 700
38.100 (1 1/2)	9.525 (3/8)	1.588 (1/16)	4.762 (3/16)	2.381 (3/32)	500 (20)	26.988 (1 1/16)	162	28 200	40 100
38.100 (1 1/2)	9.525 (3/8)	1.588 (1/16)	4.762 (3/16)	2.381 (3/32)	500 (20)	26.988 (1 1/16)	162	28 200	40 100
44.450 (1 3/4)	11.112 (7/16)	1.588 (1/16)	4.762 (3/16)	2.381 (3/32)	500 (20)	32.941 (1 19/64)	258	35 300	55 600
44.450 (1 3/4)	11.112 (7/16)	1.588 (1/16)	4.762 (3/16)	2.381 (3/32)	500 (20)	32.941 (1 19/64)	258	35 300	55 600
50.800 (2)	12.700 (1/2)	1.588 (1/16)	4.762 (3/16)	3.175 (1/8)	600 (24)	37.306 (1 15/32)	356	45 700	80 600
50.800 (2)	12.700 (1/2)	1.588 (1/16)	4.762 (3/16)	3.175 (1/8)	600 (24)	37.306 (1 15/32)	356	45 700	80 600
57.150 (2 1/4)	14.288 (9/16)	1.588 (1/16)	4.762 (3/16)	3.175 (1/8)	760 (30)	40.878 (1 39/64)	500	61 400	116 000
57.150 (2 1/4)	14.288 (9/16)	1.588 (1/16)	4.762 (3/16)	3.175 (1/8)	760 (30)	40.878 (1 39/64)	500	61 400	116 000
63.500 (2 1/2)	15.875 (5/8)	1.588 (1/16)	6.350 (1/4)	3.175 (1/8)	760 (30)	51.991 (2 3/64)	892	77 600	172 000
63.500 (2 1/2)	15.875 (5/8)	1.588 (1/16)	6.350 (1/4)	3.175 (1/8)	760 (30)	51.991 (2 3/64)	892	77 600	172 000
69.850 (2 3/4)	17.462 (11/16)	1.588 (1/16)	6.350 (1/4)	3.175 (1/8)	760 (30)	59.928 (2 23/64)	1 450	111 000	239 000
88.900 (3 1/2)	19.050 (3/4)	1.588 (1/16)	6.350 (1/4)	3.175 (1/8)	760 (30)	64.691 (2 35/64)	2 190	142 000	317 000

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

CAM FOLLOWERS

Inch Series Heavy Duty Cam Followers Full Complement Type/With Screwdriver Slot



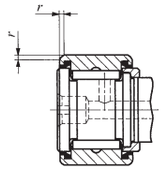


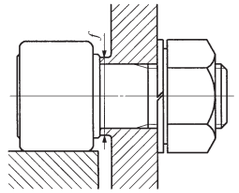
Stud dia. 6.350 — 50.800mm

CRH···V

Stud dia. mm (inch)	Identification number Shield type	Identification number Sealed type	Mass (Ref.) g	D	C	d_1	G UNF	G_1	B max
6.350 (1/4)	CRH 8-1 V	CRH 8-1 VUU	12	12.700 (1/2)	9.525 (3/8)	6.350 (1/4)	1/4 - 28	6.350 (1/4)	11.1(0.44)
	CRH 9 V	CRH 9 VUU	15	14.228 (9/16)	9.525 (3/8)	6.350 (1/4)	1/4 - 28	6.350 (1/4)	11.1(0.44)
7.938 (5/16)	CRH 10-1 V	CRH 10-1 VUU	23	15.875 (5/8)	11.112 (7/16)	7.938 (5/16)	5/16 - 24	7.938 (5/16)	12.8(0.50)
	CRH 11 V	CRH 11 VUU	27	17.462 (11/16)	11.112 (7/16)	7.938 (5/16)	5/16 - 24	7.938 (5/16)	12.8(0.50)
11.112 (7/16)	CRH 12 V	CRH 12 VUU	39	19.050 (3/4)	12.700 (1/2)	11.112 (7/16)	7/16 - 20	9.525 (3/8)	14.6(0.57)
	CRH 14 V	CRH 14 VUU	49	22.225 (7/8)	12.700 (1/2)	11.112 (7/16)	7/16 - 20	9.525 (3/8)	14.6(0.57)
15.875 (5/8)	CRH 16 V	CRH 16 VUU	93	25.400 (1)	15.875 (5/8)	15.875 (5/8)	5/8 - 18	12.700 (1/2)	17.9(0.70)
	CRH 18 V	CRH 18 VUU	109	28.575 (1 1/8)	15.875 (5/8)	15.875 (5/8)	5/8 - 18	12.700 (1/2)	17.9(0.70)
19.050 (3/4)	CRH 20 V	CRH 20 VUU	176	31.750 (1 1/4)	19.050 (3/4)	19.050 (3/4)	3/4 - 16	15.875 (5/8)	21.0(0.83)
	CRH 22 V	CRH 22 VUU	200	34.925 (1 3/8)	19.050 (3/4)	19.050 (3/4)	3/4 - 16	15.875 (5/8)	21.0(0.83)
22.225 (7/8)	CRH 24 V	CRH 24 VUU	296	38.100 (1 1/2)	22.225 (7/8)	22.225 (7/8)	7/8 - 14	19.050 (3/4)	24.3(0.96)
	CRH 26 V	CRH 26 VUU	329	41.275 (1 5/8)	22.225 (7/8)	22.225 (7/8)	7/8 - 14	19.050 (3/4)	24.3(0.96)
25.400 (1)	CRH 28 V	CRH 28 VUU	463	44.450 (1 3/4)	25.400 (1)	25.400 (1)	1 - 14 UNS	22.225 (7/8)	27.4(1.08)
	CRH 30 V	CRH 30 VUU	508	47.625 (1 7/8)	25.400 (1)	25.400 (1)	1 - 14 UNS	22.225 (7/8)	27.4(1.08)
28.575 (1 1/8)	CRH 32 V	CRH 32 VUU	722	50.800 (2)	31.750 (1 1/4)	28.575 (1 1/8)	1 1/8 - 12	25.400 (1)	34.2(1.35)
	CRH 36 V	CRH 36 VUU	858	57.150 (2 1/4)	31.750 (1 1/4)	28.575 (1 1/8)	1 1/8 - 12	25.400 (1)	34.2(1.35)
31.750 (1 1/4)	CRH 40 V	CRH 40 VUU	1 260	63.500 (2 1/2)	38.100 (1 1/2)	31.750 (1 1/4)	1 1/4 - 12	28.575 (1 1/8)	40.0(1.57)
	CRH 44 V	CRH 44 VUU	1 460	69.850 (2 3/4)	38.100 (1 1/2)	31.750 (1 1/4)	1 1/4 - 12	28.575 (1 1/8)	40.0(1.57)
38.100 (1 1/2)	CRH 48 V	CRH 48 VUU	2 100	76.200 (3)	44.450 (1 3/4)	38.100 (1 1/2)	1 1/2 - 12	31.750 (1 1/4)	46.4(1.83)
	CRH 52 V	CRH 52 VUU	2 380	82.550 (3 1/4)	44.450 (1 3/4)	38.100 (1 1/2)	1 1/2 - 12	31.750 (1 1/4)	46.4(1.83)
44.450 (1 3/4)	CRH 56 V	CRH 56 VUU	3 240	88.900 (3 1/2)	50.800 (2)	44.450 (1 3/4)	1 3/4 - 12 UN	34.925 (1 3/8)	52.8(2.08)
50.800 (2)	CRH 64 V	CRH 64 VUU	4 960	101.600 (4)	57.150 (2 1/4)	50.800 (2)	2- 12 UN	38.100 (1 1/2)	59.4(2.34)

Remarks1. Models with a stud diameter d_1 of 7.938 mm or less (marked *) are provided with an oil hole on the stud head only. Other models are provided with one oil hole each on the head, outside surface and end surface of the stud.
2. Provided with prepacked grease.





CRH···VUU

Boundary dimensions mm(inch) B_2	B_3	C_1	g_1	g_2	r	Mounting dimension f Min. mm(inch)	Maximum tightening torque N-m	Basic dynamic load rating C N	Basic static load rating C_0 N
15.875 (5/8)	— (—)	0.794 (1/32)	*3.175 (1/8)	— (—)	0.397 (1/64)	8.334 (21/64)	3.4	4 710	5 410
15.875 (5/8)	— (—)	0.794 (1/32)	*3.175 (1/8)	— (—)	0.397 (1/64)	8.334 (21/64)	3.4	4 710	5 410
19.050 (3/4)	— (—)	0.794 (1/32)	*3.175 (1/8)	— (—)	0.397 (1/64)	11.112 (7/16)	6.8	6 340	8 530
19.050 (3/4)	— (—)	0.794 (1/32)	*3.175 (1/8)	— (—)	0.397 (1/64)	11.112 (7/16)	6.8	6 340	8 530
22.225 (7/8)	6.350 (1/4)	0.794 (1/32)	4.762 (3/16)	2.381 (3/32)	0.794 (1/32)	13.494 (17/32)	17.6	8 710	12 300
22.225 (7/8)	6.350 (1/4)	0.794 (1/32)	4.762 (3/16)	2.381 (3/32)	0.794 (1/32)	13.494 (17/32)	17.6	8 710	12 300
25.400 (1)	6.350 (1/4)	1.588 (1/16)	4.762 (3/16)	2.381 (3/32)	1.191 (3/64)	18.256 (23/32)	57.8	13 100	22 700
25.400 (1)	6.350 (1/4)	1.588 (1/16)	4.762 (3/16)	2.381 (3/32)	1.588 (1/16)	18.256 (23/32)	57.8	13 100	22 700
31.750 (1 1/4)	7.938 (5/16)	1.588 (1/16)	4.762 (3/16)	2.381 (3/32)	1.588 (1/16)	24.209 (61/64)	103	23 600	31 700
31.750 (1 1/4)	7.938 (5/16)	1.588 (1/16)	4.762 (3/16)	2.381 (3/32)	1.588 (1/16)	24.209 (61/64)	103	23 600	31 700
38.100 (1 1/2)	9.525 (3/8)	1.588 (1/16)	4.762 (3/16)	2.381 (3/32)	1.588 (1/16)	26.988 (1 1/16)	162	28 200	40 100
38.100 (1 1/2)	9.525 (3/8)	1.588 (1/16)	4.762 (3/16)	2.381 (3/32)	1.588 (1/16)	26.988 (1 1/16)	162	28 200	40 100
44.450 (1 3/4)	11.112 (7/16)	1.588 (1/16)	4.762 (3/16)	2.381 (3/32)	1.588 (1/16)	32.941 (1 19/64)	258	35 300	55 600
44.450 (1 3/4)	11.112 (7/16)	1.588 (1/16)	4.762 (3/16)	2.381 (3/32)	1.588 (1/16)	32.941 (1 19/64)	258	35 300	55 600
50.800 (2)	12.700 (1/2)	1.588 (1/16)	4.762 (3/16)	3.175 (1/8)	1.588 (1/16)	37.306 (1 15/32)	356	45 700	80 600
50.800 (2)	12.700 (1/2)	1.588 (1/16)	4.762 (3/16)	3.175 (1/8)	1.588 (1/16)	37.306 (1 15/32)	356	45 700	80 600
57.150 (2 1/4)	14.288 (9/16)	1.588 (1/16)	4.762 (3/16)	3.175 (1/8)	2.381 (3/32)	40.878 (1 39/64)	500	61 400	116 000
57.150 (2 1/4)	14.288 (9/16)	1.588 (1/16)	4.762 (3/16)	3.175 (1/8)	2.381 (3/32)	40.878 (1 39/64)	500	61 400	116 000
63.500 (2 1/2)	15.875 (5/8)	1.588 (1/16)	6.350 (1/4)	3.175 (1/8)	2.381 (3/32)	51.991 (2 3/64)	892	77 600	172 000
63.500 (2 1/2)	15.875 (5/8)	1.588 (1/16)	6.350 (1/4)	3.175 (1/8)	2.381 (3/32)	51.991 (2 3/64)	892	77 600	172 000
69.850 (2 3/4)	17.462 (11/16)	1.588 (1/16)	6.350 (1/4)	3.175 (1/8)	2.381 (3/32)	59.928 (2 23/64)	1 450	111 000	239 000
88.900 (3 1/2)	19.050 (3/4)	1.588 (1/16)	6.350 (1/4)	3.175 (1/8)	2.381 (3/32)	64.691 (2 25/64)	2 190	142 000	317 000

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

ROLLER FOLLOWERS

- Separable Roller Followers
- Non-separable Roller Followers
- Heavy Duty Type Roller Followers



Structure and Features

IKO Roller Followers are bearings designed for outer ring rotation, in which needle rollers are incorporated in a thick walled outer ring. Both crowned and cylindrical outer rings are available. The outer rings run directly on mating track surfaces, and the crowned outer ring is effective in relieving the edge load caused by mounting errors. The cylindrical outer ring, on the other hand, has a large contact area with the mating track surface and is suitable for applications involving large loads or low track surface hardness.

In Roller Followers, there are two types of bearings available, the caged type and the full complement type. The caged type is useful for applications at high-speed rotation. The full complement type, on the other hand, is suitable for heavy-load applications at low-speed rotation or oscillating motions.

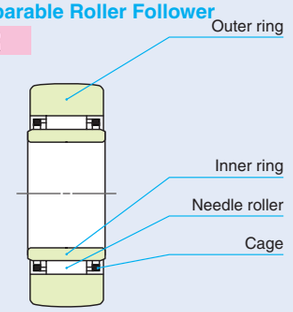
Roller Followers include separable and non-separable types. Also, in addition to the open type, shield type and sealed type are available. The clearances between the side plates and outer ring of the shield type are narrow, and form labyrinths. In the sealed type, special synthetic rubber seals are assembled in these clearances, and they are effective in preventing penetration of dust and dirt.

These bearings are available in a variety of types to suit almost any kind of application. They are widely used for cam mechanisms and for linear motions of conveying equipment.

Structures of Roller Followers

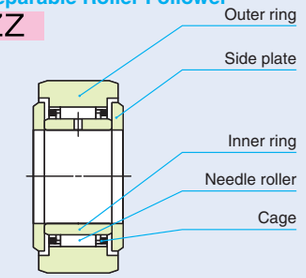
Structure of Separable Roller Follower

NAST···R



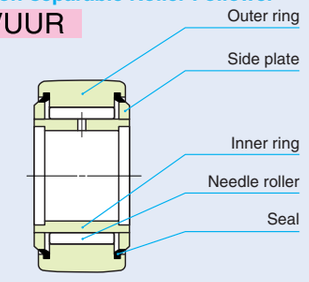
Structure of Separable Roller Follower

NAST···ZZ



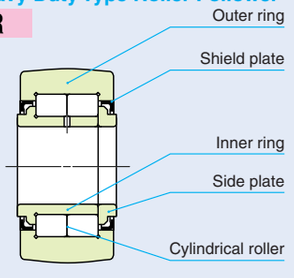
Structure of Non-separable Roller Follower

NART···VUUR



Structure of Heavy Duty Type Roller Follower

NURT···R



Types

In Roller Followers, types shown in Table 1 are available.

Table 1 Type of Roller Followers

Type				With cage		Full complement type	
				Crowned outer ring	Cylindrical outer ring	Crowned outer ring	Cylindrical outer ring
Metric series	Separable Roller Followers RNAST, NAST	Without inner ring	Open type	RNAST··· R	RNAST	—	—
		With inner ring	Open type	NAST··· R	NAST	—	—
			Shield type	NAST···ZZ R	NAST···ZZ	—	—
			Sealed type	NAST···ZZUUR	NAST···ZZUU	—	—
	Non-separable Roller Followers NART		Shield type	NART··· R	—	NART···V R	—
			Sealed type	NART··· UUR	—	NART···VUUR	—
	Heavy Duty Type Roller Followers NURT		Shield type	—	—	NURT··· R	NURT
Inch series	Non-separable Roller Followers CRY		Shield type	—	—	CRY ···V R	CRY ···V
			Sealed type	—	—	CRY ···VUUR	CRY···VUU

Separable Roller Followers

These bearings are assembled by combining an outer ring, inner ring and Needle Roller Cage, which can be separated from one another. Thus, handling is easy. Oil lubrication is also easy, making them suitable for high-speed rotations.
There are two types: type without inner ring RNAST and type with inner ring NAST. The type with inner ring includes open type, shield type, and sealed type.

Non-separable Roller Followers

These non-separable type bearings have side plates fixed on both sides of the inner ring, and include the caged type and the full complement type. Both shield type and sealed type are available.
Inch series Non-separable Roller Followers are full complement type bearings and their surface is treated with black oxide surface treatment.

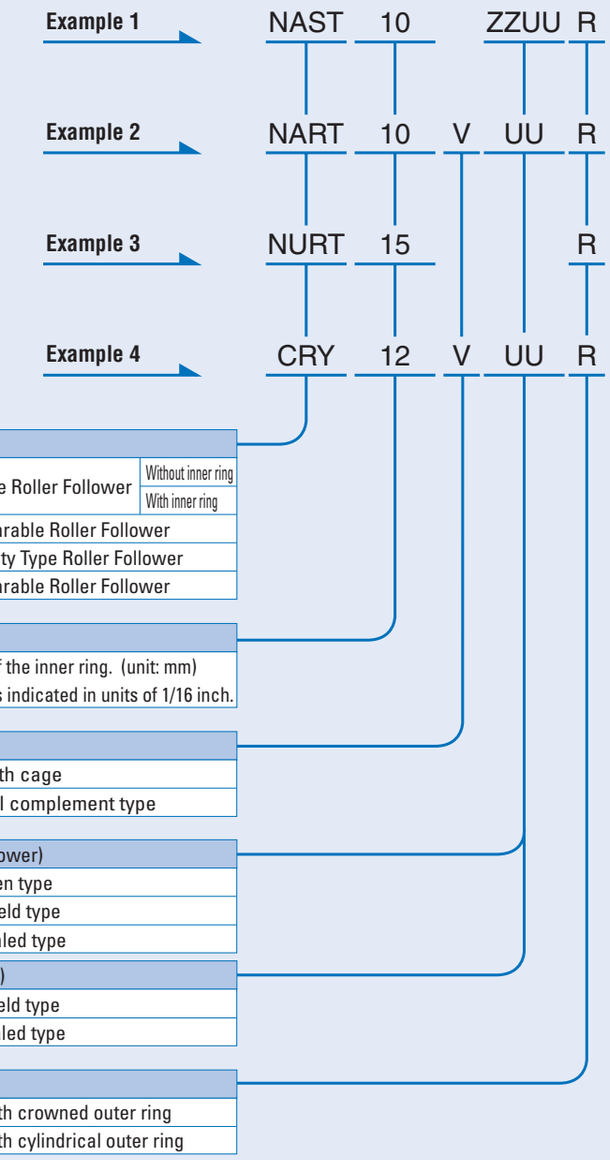
Heavy Duty Type Roller Followers

These full complement type bearings incorporate cylindrical rollers in the outer ring in two rows and can withstand large radial loads and some axial loads. These bearings are shield type with non-separable structure.

Identification Number

Some examples of the identification number of Roller Followers are shown below.

Examples of identification number



Model code		
Metric series	RNAST	Separable Roller Follower (Without inner ring)
	NAST	Separable Roller Follower (With inner ring)
	NART	Non-separable Roller Follower
	NURT	Heavy Duty Type Roller Follower
Inch series	CRY	Non-separable Roller Follower

Size
The size indicates the bore diameter of the inner ring. (unit: mm)
In the inch series, the bore diameter is indicated in units of 1/16 inch.

Roller guide method	
No symbol	With cage
V	Full complement type

Seal structure (Separable Roller Follower)	
No symbol	Open type
ZZ	Shield type
ZZUU	Sealed type

Seal structure (Other Roller Follower)	
No symbol	Shield type
UU	Sealed type

Shape of outer ring outside surface	
R	With crowned outer ring
No symbol	With cylindrical outer ring

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

Accuracy

Dimensional accuracy and rotational accuracy of Roller Followers are based on Tables 2, 3 and 4. Tolerances for the smallest single roller set bore diameter of Separable Roller Followers are shown in Table 5. Roller Followers with special accuracy can also be manufactured. Please contact IKO.

Table 2 Tolerances

unit: μm

Dimensions and symbols \ Series		Metric series		Inch series	
		Crowned outer ring	Cylindrical outer ring	Crowned outer ring	Cylindrical outer ring
Bore dia. of inner ring d	$d \leqq 19.05$	See Table 3.		+5 −10	+5 −10
	$19.05 < d$			+2 −12	
Outside dia. of outer ring D		0 −50	See Table 4.	0 −50	0 −25
Width of outer ring C		0 −120		0 −130	
Width of inner ring B	Separable Roller Follower	0 −120		—	
Width of bearing B	Non-separable Roller Follower	h12	—	+130 −250	
	Heavy Duty Type Roller Follower				
Roller set bore dia. F_w	Separable Roller Follower	See Table 5.		—	

Table 3 Tolerances and allowable values of inner rings (Metric series)

unit: μm

d Nominal bore dia. mm		Δ_{dmp} Single plane mean bore dia. deviation		V_{dp} Bore dia. variation in a single radial plane	V_{dmp} Mean bore dia. variation	K_{ia} Radial runout of assembled bearing inner ring	V_{Bs} Width variation
Over	Incl.	High	Low	(Max.)	(Max.)	(Max.)	(Max.)
2.5	10	0	−8	10	6	10	15
10	18	0	−8	10	6	10	20
18	30	0	−10	13	8	13	20
30	50	0	−12	15	9	15	20

Table 4 Tolerances and allowable values of outer rings (Metric series)

unit: μm

D Nominal outside dia. of outer ring mm		Δ_{Dmp} Single plane mean outside dia. deviation		V_{Dp} (1) Outside dia. variation in a single radial plane	V_{Dmp} (1) Mean outside dia. variation	K_{ea} (1) Radial runout of assembled bearing outer ring	V_{Cs} Width variation
Over	Incl.	High	Low	(Max.)	(Max.)	(Max.)	(Max.)
6	18	0	−8	10	6	15	Same as the tolerance values of V_{Bs} for d of the inner of the same bearing
18	30	0	−9	12	7	15	
30	50	0	−11	14	8	20	
50	80	0	−13	16	10	25	
80	120	0	−15	19	11	35	

Note(1) Also applicable to the inch series.

Table 5 Tolerances of smallest single roller set bore diameter $F_{ws\,min}$

unit: μm

F_w Nominal roller set bore diameter mm		$\Delta_{Fws\,min}$ Deviation of smallest single roller set bore diameter	
Over	Incl.	High	Low
6	10	+22	+13
10	18	+27	+16
18	30	+33	+20
30	50	+41	+25
50	80	+49	+30

Clearance

Radial internal clearances of Roller Followers are based on Table 6.

Table 6 Radial internal clearance

unit: μm

Identification number (1)				Radial internal clearance	
Metric series			Inch series		
Separable Roller Followers	Non-separable Roller Followers	Heavy Duty Type Roller Followers	Non-separable Roller Followers	Min.	Max.
NAST 6R	NART 5R	—	—	5	20
NAST 8R～NAST12R	NART 6R～NART12R	—	—	5	25
NAST15R～NAST25R	NART15R～NART20R	—	—	10	30
NAST30R～NAST40R	NART25R～NART40R	—	—	10	40
NAST45R, NAST50R	NART45R, NART50R	—	—	15	50
—	—	NURT15R～NURT30-1R	—	20	45
—	—	NURT35R～NURT40-1R	—	25	50
—	—	NURT45R～NURT50-1R	—	30	60
—	—	—	CRY12R～CRY56R	35	60
—	—	—	CRY64R	45	70

Note(1) Also applicable to the full complement type, cylindrical outer ring type, shield type and sealed type.

Fit

Roller Followers are generally used under the loading conditions in which the load direction is fixed in relation to the inner ring and rotates in relation to the outer ring. The recommended fits for shafts are shown in Table 7. Those for the inch series are shown in the dimension table.

Table 7 Recommended fit (Metric series)

Type		Tolerance class of shaft
Separable Roller Followers	without inner ring	k5, k6
	with inner ring	g6, h6
Non-separable Roller Followers		
Heavy Duty Type Roller Followers		

Maximum allowable static load

The load that is applicable to Roller Followers is, in some cases, determined by the strength of the outer ring rather than by the load rating of the needle roller bearing. Therefore, the maximum allowable load that is limited by the strength of outer ring is specified.

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

Track Capacity

Track capacity is defined as the load that can be continuously applied on a Roller Follower placed on a steel track surface without causing deformation and indentation on the track surface when the outer ring of the Roller Follower makes contact with the mating track surface (plane). The track capacities shown in Tables 8.1 and 8.2 are applicable when the hardness of the mating track surface is 40HRC (Tensile strength 1250N/mm²). When the hardness of the mating track surface differs from 40HRC, the track capacity is obtained by multiplying the value by the track capacity factor shown in Table 9.

If lubrication between the outer ring and the mating track surface is insufficient, seizure and/or wear may occur depending on the application. Therefore, pay attention to lubrication and surface roughness of the mating track especially in the case of high-speed rotation such as for cam mechanisms.

Table 8.1 Track capacity (Metric series) unit: N

Roller Followers with crowned outer ring				Roller Followers with cylindrical outer ring					
Identification number (1)			Track capacity	Identification number	Track capacity	Identification number (2)	Track capacity	Identification number	Track capacity
Separable Roller Followers	Non-separable Roller Followers	Heavy Duty Type Roller Followers							
RNAST 5R	NART 5R	—	1 040	RNAST 5	2 310	—	—	—	—
(R)NAST 6R	NART 6R	—	1 330	(R)NAST 6	3 550	NAST 6ZZ	3 550	—	—
(R)NAST 8R	NART 8R	—	1 850	(R)NAST 8	3 980	NAST 8ZZ	4 490	—	—
(R)NAST10R	NART10R	—	2 470	(R)NAST10	5 610	NAST10ZZ	6 890	—	—
(R)NAST12R	NART12R	—	2 710	(R)NAST12	5 990	NAST12ZZ	7 350	—	—
(R)NAST15R	NART15R	NURT15 R	3 060	(R)NAST15	6 550	NAST15ZZ	8 030	NURT15	11 500
—	—	NURT15-1R	3 910	—	—	—	—	NURT15-1	13 700
(R)NAST17R	NART17R	NURT17 R	3 660	(R)NAST17	10 900	NAST17ZZ	11 700	NURT17	13 600
—	—	NURT17-1R	4 530	—	—	—	—	NURT17-1	16 000
(R)NAST20R	NART20R	NURT20 R	4 530	(R)NAST20	12 800	NAST20ZZ	13 800	NURT20	20 000
—	—	NURT20-1R	5 190	—	—	—	—	NURT20-1	22 100
(R)NAST25R	NART25R	NURT25 R	5 190	(R)NAST25	14 100	NAST25ZZ	15 300	NURT25	22 100
—	—	NURT25-1R	6 580	—	—	—	—	NURT25-1	26 400
(R)NAST30R	NART30R	NURT30 R	6 580	(R)NAST30	22 100	NAST30ZZ	22 100	NURT30	31 600
—	—	NURT30-1R	8 020	—	—	—	—	NURT30-1	36 700
(R)NAST35R	NART35R	NURT35 R	8 020	(R)NAST35	25 700	NAST35ZZ	25 700	NURT35	36 700
—	—	NURT35-1R	9 220	—	—	—	—	NURT35-1	40 800
(R)NAST40R	NART40R	NURT40 R	9 220	(R)NAST40	26 900	NAST40ZZ	30 300	NURT40	44 200
—	—	NURT40-1R	10 800	—	—	—	—	NURT40-1	49 700
(R)NAST45R	NART45R	NURT45 R	9 990	(R)NAST45	28 500	NAST45ZZ	32 200	NURT45	47 000
—	—	NURT45-1R	12 400	—	—	—	—	NURT45-1	55 300
(R)NAST50R	NART50R	NURT50 R	10 800	(R)NAST50	30 200	NAST50ZZ	34 000	NURT50	49 700
—	—	NURT50-1R	14 000	—	—	—	—	NURT50-1	60 800

Notes(1) Also applicable to the full complement type, shield type, and sealed type.
(2) Also applicable to the sealed type.

Table 8.2 Track capacity (Inch series) unit: N

Crowned outer ring		Cylindrical outer ring	
Identification number (1)	Track capacity	Identification number (1)	Track capacity
CRY12R	853	CRY12	4 490
CRY14R	1 050	CRY14	5 240
CRY16R	1 420	CRY16	7 270
CRY18R	1 660	CRY18	7 700
CRY20R	2 160	CRY20	10 700
CRY22R	2 450	CRY22	11 800
CRY24R	3 410	CRY24	15 400
CRY26R	3 820	CRY26	16 700
CRY28R	4 210	CRY28	21 000
CRY30R	4 610	CRY30	22 500
CRY32R	5 690	CRY32	30 800
CRY36R	6 640	CRY36	34 700
CRY40R	8 970	CRY40	44 900
CRY44R	10 200	CRY44	49 400
CRY48R	11 400	CRY48	64 300
CRY52R	12 700	CRY52	69 600
CRY56R	14 100	CRY56	87 000
CRY64R	16 800	CRY64	113 000

Note(1) Also applicable to the sealed type.

Table 9 Track capacity factor

Hardness HRC	Tensile strength N/mm²	Track capacity factor	
		Crowned outer ring	Cylindrical outer ring
20	760	0.22	0.37
25	840	0.31	0.46
30	950	0.45	0.58
35	1 080	0.65	0.75
38	1 180	0.85	0.89
40	1 250	1.00	1.00
42	1 340	1.23	1.15
44	1 435	1.52	1.32
46	1 530	1.85	1.51
48	1 635	2.27	1.73
50	1 760	2.80	1.99
52	1 880	3.46	2.29
54	2 015	4.21	2.61
56	2 150	5.13	2.97
58	2 290	6.26	3.39

Allowable Rotational Speed

The allowable rotational speed of Roller Followers is affected by mounting and operating conditions. For reference, Table 10 shows dn values when only pure radial loads are applied. Under actual operating conditions, the recommended dn value is 1/10 of the value shown in the table in consideration of the axial loads that may act on the bearing.

Table 10 dn values of Roller Followers(1)

Type \ Lubricant	Grease	Oil
Caged type	84 000	140 000
Full complement type	42 000	70 000
Heavy Duty Type Roller Follower	72 000	120 000

Note(1) dn value $= d \times n$
where, d : Bore diameter of bearing mm
n : Rotational speed rpm

Lubrication

In Sealed Type Roller Followers, Heavy Duty Type Roller Followers and Inch series Roller Followers, ALVANIA GREASE S2 (SHELL) is prepacked as the lubricating grease.

For Roller Followers without prepacked grease, grease or oil should be supplied through the oil hole of the inner ring for use. If they are used without lubrication, wear of rolling contact surfaces may take place, leading to a short bearing life.

Oil Hole

Open Type Separable Roller Followers have no oil hole. Inner rings of other types of Metric series Roller Followers have an oil hole. Inch series inner rings have an oil groove and an oil hole.

Mounting

❶In case of shield and sealed types, match the side surface correctly to the mating seating surface indicated by the dimension a shown in the dimension table, and fix them. (See Fig. 1.)

❷When mounting Roller Followers, pay special attention to avoid locating the oil hole of the inner ring within the loading zone. This may lead to a short bearing life. (See Fig. 2.)

❸When mounting Sealed Type Separable Roller Followers, do not cause the side plates to come off. If they come off, set them again in place taking care to avoid damaging the seal lips.

❹In case of Roller Followers without an inner ring, the shaft requires heat treatment and grinding finish. The recommended surface hardness of the shaft is 58～64HRC, and the recommended roughness of the shaft is 0.2 μmR_a or less.
Also, the outer ring and cage are guided by side surfaces of the mounting parts. Therefore, it is recommended that the side surfaces of the mounting parts be finished by grinding or at least by machining. (See Fig. 3.)

❺In Non-separable Roller Followers, the side plates are press-fitted. Therefore, when mounting the Roller Followers, do not push the side plates.

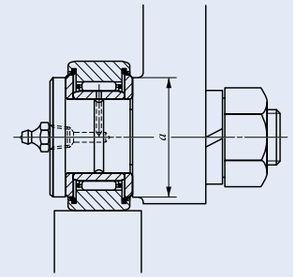


Fig. 1 Mating seating dimension "a"

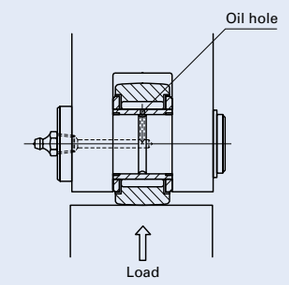


Fig. 2 Position of oil hole and load direction

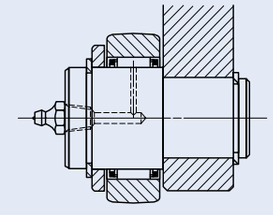


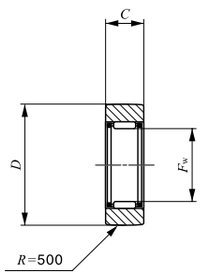
Fig. 3 Mounting example of Roller Follower without inner ring

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

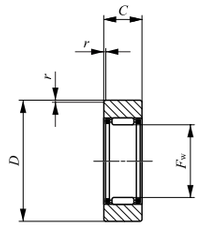
ROLLER FOLLOWERS

Separable Roller Followers, Open Type With Cage/Without Inner Ring





RNAST···R



RNAST

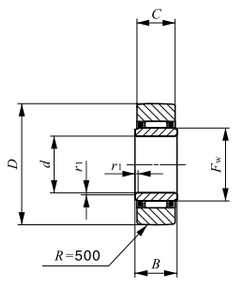
Shaft dia. 7 — 60mm

Shaft dia. mm	Identification number Open type: Crowned outer ring	Open type: Cylindrical outer ring	Mass (Ref.) g	Boundary dimensions mm: F_w	D	C	$r_{s\,min}$ (1)	Basic dynamic load rating C N	Basic static load rating C_0 N
7	RNAST 5 R	RNAST 5	8.9	7	16	7.8	0.3	2 710	2 390
10	RNAST 6 R	RNAST 6	13.9	10	19	9.8	0.3	4 160	4 550
12	RNAST 8 R	RNAST 8	23.5	12	24	9.8	0.6	5 650	5 890
14	RNAST 10 R	RNAST 10	42.5	14	30	11.8	1	9 790	9 680
16	RNAST 12 R	RNAST 12	49.5	16	32	11.8	1	10 500	10 900
20	RNAST 15 R	RNAST 15	50	20	35	11.8	1	12 400	14 300
22	RNAST 17 R	RNAST 17	90	22	40	15.8	1	17 600	20 900
25	RNAST 20 R	RNAST 20	135	25	47	15.8	1	19 400	24 500
30	RNAST 25 R	RNAST 25	152	30	52	15.8	1	20 800	28 400
38	RNAST 30 R	RNAST 30	255	38	62	19.8	1	30 500	45 400
42	RNAST 35 R	RNAST 35	375	42	72	19.8	1	32 400	50 600
50	RNAST 40 R	RNAST 40	420	50	80	19.8	1.5	35 900	61 100
55	RNAST 45 R	RNAST 45	460	55	85	19.8	1.5	37 400	66 400
60	RNAST 50 R	RNAST 50	500	60	90	19.8	1.5	38 900	71 700

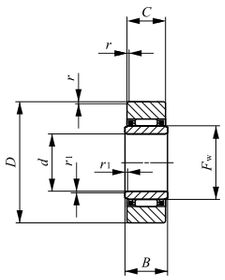
Note(1) Minimum allowable value of chamfer dimension r
Remarks1. No oil hole is provided.
2. Not provided with prepacked grease. Perform proper lubrication for use.

Separable Roller Followers, Open Type With Cage/With Inner Ring





NAST···R



NAST

Shaft dia. 6 — 50mm

Shaft dia. mm	Identification number Open type: Crowned outer ring	Open type: Cylindrical outer ring	Mass (Ref.) g	Boundary dimensions mm: d	D	B	C	$r_{s\,min}$ (1)	$r_{1s\,min}$ (1)	F_w	Basic dynamic load rating C N	Basic static load rating C_0 N	Assembled inner ring
6	NAST 6 R	NAST 6	17.8	6	19	10	9.8	0.3	0.3	10	4 160	4 550	LRT 61010 S
8	NAST 8 R	NAST 8	28	8	24	10	9.8	0.6	0.3	12	5 650	5 890	LRT 81210 S
10	NAST 10 R	NAST 10	49.5	10	30	12	11.8	1	0.3	14	9 790	9 680	LRT 101412 S
12	NAST 12 R	NAST 12	58	12	32	12	11.8	1	0.3	16	10 500	10 900	LRT 121612 S
15	NAST 15 R	NAST 15	62	15	35	12	11.8	1	0.3	20	12 400	14 300	LRT 152012 S
17	NAST 17 R	NAST 17	109	17	40	16	15.8	1	0.3	22	17 600	20 900	LRT 172216 S
20	NAST 20 R	NAST 20	157	20	47	16	15.8	1	0.3	25	19 400	24 500	LRT 202516 S
25	NAST 25 R	NAST 25	180	25	52	16	15.8	1	0.3	30	20 800	28 400	LRT 253016 S
30	NAST 30 R	NAST 30	320	30	62	20	19.8	1	0.6	38	30 500	45 400	LRT 303820 S
35	NAST 35 R	NAST 35	440	35	72	20	19.8	1	0.6	42	32 400	50 600	LRT 354220 S
40	NAST 40 R	NAST 40	530	40	80	20	19.8	1.5	1	50	35 900	61 100	LRT 405020 S
45	NAST 45 R	NAST 45	580	45	85	20	19.8	1.5	1	55	37 400	66 400	LRT 455520 S
50	NAST 50 R	NAST 50	635	50	90	20	19.8	1.5	1	60	38 900	71 700	LRT 506020 S

Note(1) Minimum allowable value of chamfer dimension r or r_1
Remarks1. No oil hole is provided.
2. Not provided with prepacked grease. Perform proper lubrication for use.

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch